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Nickel recovery from Ni-MH batteries using H₂SO₄ leaching.

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1 . Introduction

In recent years, the escalating demand for batteries and portable electronic devices has led to a significant surge in the production and consumption of rechargeable batteries. Among the various materials used in these batteries, nickel plays a crucial role as a key component of the battery's cathode. As the lifespan of batteries diminishes and the need for sustainable resource management grows, there is a pressing need for efficient and environmentally friendly methods to recover valuable metals like nickel and cobalt from spent batteries.

One promising and widely researched approach for nickel recovery is the use of sulfuric acid (H_2SO_4) leaching. This method involves the selective dissolution of nickel from the cathode materials, leaving behind other components. The process is not only environmentally sustainable but also economically viable, as it allows for the recycling of valuable materials, reducing the reliance on traditional mining and minimizing the environmental impact associated with raw material extraction.

This introductory exploration into nickel recovery from batteries using H_2SO_4 leaching aims to delve into the principles, mechanisms, and potential advantages of this innovative process. By understanding the science behind the leaching method, we can develop efficient strategies for recovering nickel and other valuable metals from spent batteries, contributing to a more circular and sustainable approach to battery production and waste management.

So, we decided to use an AA- and AAA-type cylindrical spent nickel-metal hydride (Ni-MH) batteries in our plan for nickel recovery because it's contained a high percent of nickel in the cathode and have a simple construction that helps to doing the nickel recovery without investing a lot of processes.

1.2 The purpose of the project

Design a plan for nickel recovery from Ni-MH batteries using H_2SO_4 leaching.

1.3 Quantitative basis

Purity: 99.6 %.⁹

Worldwide mine production of nickel ^[11]: 3,300,000 tons in 2022.^[21]

Production days: 300 days (~7200 hours).

Operation: Continuous process.

The price of 99.6% purity nickel per ton varies between 16,000-30000 USD, depending on the amount purchased, quality, & composition of the impurities in the nickel ^[12] ^[21].

As for NiMH battery production:

there are not specific studies and statistics regarding the available recycled NiMH batteries we can roughly estimate.

if we do a quick estimation:

*We will refer to this (-) as a range

Global NiMH battery market size in 2022: USD 3.34 billion.^{[14][21]}

Asia pacific share of global NiMH production 40-45%.^[13]

3.34 billion *(0.4-0.45) =(1.3b-1.5b)

The estimated average price per unit is 5 \$.^[15]

(1.3b-1.5b)/5=(260M-300M)

50%-60% of NiMH are recycled globally .^{[16] [21]}

(260M-300M) *(0.5-0.6) =130M-180M

Add to that the 30% scrape production.^[17]

(1.3b-1.5b) =70%

(1.8b-2.1b) *0.3=540M-630M

(130M-180M) +(540M-630M) =670M-810M batteries

15-25 grams average wight.^[18] = (1.6-2.7*10⁻⁵) ton

(670M-810M) * ((1.6-2.7) *10⁻⁵) = (10720 -21870) ton

The estimated quantity of available NiMH batteries for recycling in Asia pacific is 11,000 to 22,000 tons per year.

Japan is the leading producer in Asia pacific in NiMH batteries and if we estimate that it produces more than half the batteries in Aisa pacific.^[19], we can roughly estimate that more than 6,000-11,000 tons would be available to recycle in a year in Japan.

$$\left(\frac{6,000 \frac{\text{ton}}{\text{year}}}{300 \text{ day}} \right) - \left(\frac{11,000 \frac{\text{ton}}{\text{year}}}{300 \text{ day}} \right) = 20 \frac{\text{ton}}{\text{day}} - 37 \frac{\text{ton}}{\text{day}} = 834 \frac{\text{Kg}}{\text{hr}} - 1542 \frac{\text{Kg}}{\text{hr}}$$

1.4 Main uses of nickel

- 1.** Battery Manufacturing: Nickel hydroxide is a key component in the production of rechargeable batteries, such as nickel-metal hydride (NiMH) and nickel-cadmium (Ni-Cd) batteries.
- 2.** Electroplating: Nickel hydroxide can be processed to produce nickel metal, which is used in electroplating applications to provide a protective and decorative coating on surfaces.
- 3.** Catalysts: Nickel-based catalysts find applications in chemical processes, including hydrogenation reactions in the manufacturing of various chemicals.
- 4.** Alloys: Nickel is an essential component in the production of alloys, such as stainless steel, which is widely used in the manufacturing of appliances, cutlery, and construction materials.
- 5.** Electronics: Nickel is used in various electronic components, including connectors, switches, and integrated circuits.
- 6.** Aerospace Industry: Nickel-based alloys are employed in aerospace applications due to their high strength and resistance to corrosion at high temperatures.
- 7.** Medical Devices: Nickel alloys are used in the production of medical devices and implants due to their biocompatibility and corrosion resistance.

2.1 Nickel recovery from Ni-MH batteries using sulfuric acid leaching.

Nickel recovery process from spent AA- and AAA-type cylindrical nickel-metal hydride (Ni-MH) batteries, sourced from Japan and collected from local scraps, meticulous attention to the initial stages of electrode material preparation. Following the discharging of the batteries, the disassembly of batteries in a scutter cutter with deionized water as the medium takes place, heavier particles, such as the steel casing, are efficiently separated from the electrode material. The lighter nylon separator is floated away, and the remaining black powder undergoes wet screening. The electrode powder, approximately 73.17 g, undergoes thorough washing repeatedly with deionized water to eliminate potential metal contaminants, particularly sodium and potassium. This rigorous washing prevents these metals from acting as activators in subsequent chemical reactions. Following the washing step, the battery powder undergoes meticulous drying at 80°C to eliminate any residual moisture.

Subsequently, calcination at 550°C is employed to remove organic matter, emphasizing a commitment to material purity and quality. The sulfuric acid leaching of metals from the electrode material unfolds in a controlled 500 mL three-necked round-bottom flask equipped with a magnetic stirrer, sensor-based temperature controller, and a reflux condenser. Maintaining stirring conditions at an appropriate speed of 500 rpm minimizes diffusion layers. Continuous in-line monitoring, facilitated by cutting-edge automated systems, ensures real-time progress, with an automated sampling system handling regular analysis. The leaching reaction, involving $Ni(OH)_2 + H_2SO_4 \rightarrow NiSO_4 + 2H_2O$, is a cornerstone of the process. Following leaching, heating in an electric heater to evaporate most of the byproduct water and to increase the concentration of the solutes, then cooling in electric cooler for $NiSO_4$ crystallization and filtration to separate the $NiSO_4$ crystal from the solution. Metal content calculations are seamlessly automated for efficiency. The culmination of this comprehensive process yields the final product: high-quality nickel, achieved through advanced electrolysis. The electrolysis reaction, $Ni^{2+} + 2e^- \rightarrow Ni$. This high-quality nickel is meticulously packaged.^{1 2}

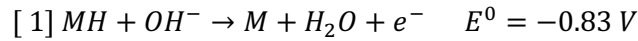
Table 1: The process

Step	Description
1	Source batteries: AA- and AAA-type Ni-MH batteries produced in Japan and collected locally from scraps.
2	Discharging the Ni-MH batteries
3	Disassembly: Ground and crushed in a scutter cutter using deionized water to remove steel case, associated battery electrode, and nylon separator.
4	Separation of components: Heavy particles settle, nylon separator floats away; black powder undergoes wet screening.
5	washing repeatedly with deionized water to eliminate potential metal contaminants from electrode powder.
6	Drying process: Battery powder dried at 80°C for 24 hours to eliminate moisture.
7	Calcination: Black powder undergoes calcination at 550°C for 3 hours to remove organic matter.
8	Leaching of metals from electrode material in a 500mL Three-necked round-bottom flask. $Ni(OH)_2 + H_2SO_4 \rightarrow NiSO_4 + 2H_2O$
9	heating in an electric heater to evaporate most of the byproduct water

10	cooling in electric cooler for $NiSO_4$ crystallization
11	filtration to separate the $NiSO_4$ crystal from the solution
12	Production of nickel through advanced electrolysis or precipitation methods. $Ni^{2+} + 2e^- \rightarrow Ni$.

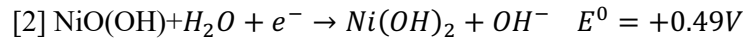
2.2 Reactions.

During discharge, the hydrogen bound in the metal hydride MH is oxidized to a proton:.^[1]

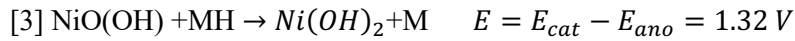


The resulting protons react with the OH^- hydroxide ions of the KOH solution to form water, The separator holds the electrolyte and prevents direct contact with the positive electrode.

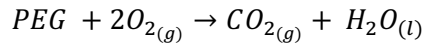
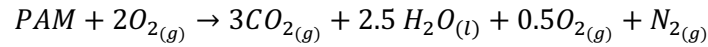
The cathode consists of a sheet of nickel hydroxide and nickel oxide hydrate:



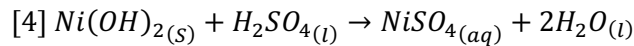
The overall reaction during discharge is:



The combustion reaction:



The H_2SO_4 leaching reaction of the base metals present in the spent battery powder materials:



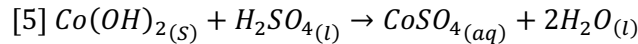
$NiSO_{4(aq)}$ – Nickel sulfate crystals is a prime product.

$H_2O_{(l)}$ – Water is a byproduct.

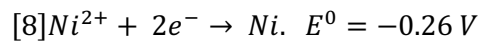
$Ni(OH)_{2(s)}$ – Nickel hydroxide, one of the materials that consist of the electrodes

The adverse reactions:

The dissolution of manganese containing phase can proceed as:



Production of nickel through advanced electrolysis, reduction reaction occurs in the cathode:



The optimum conditions comprising of 2M

H_2SO_4 , 100 $\frac{g}{L}$ pulp density and 348[K] temperature after 120 min of leaching.

2.3 Compounds enthalpy of formation.:

Table 2. Compounds enthalpy of formation.^[5,6,7]

$\Delta H_R(T, P) = \sum (\Delta H_f^\circ)_{products} - \sum (\Delta H_f^\circ)_{reactants}$	
Substance	$\Delta H_f^\circ \left(\frac{KJ}{mol} \right)$ at 298 K, 1 bar
$Ni(OH)_2$	-538.0
H_2SO_4	-909.27
$NiSO_4$	-95.8
H_2O	-285.83
$Co(OH)_2$	-539.74
$CoSO_4$	!
Ni^{+2}	-53.97
Ni	0

2.4 The heat reactions:

Table 3. the heat reactions

Number of the equation	Heat reaction $\left(\frac{KJ}{mol} \right)$	Endo/Exo
4	-779.81	Exo
5	!	!

2.5 The calculated activation energy values acquired during the leaching of the base metals:

Table 4. the activation energy reactions.^[1]

Number of the equation	Activation energy $\left[\frac{kJ}{mol} \right]$
4	8.7
5	6.8

2.6 Kinetics of the reactions.

the model who describes the kinetic of the leaching reaction.^[8,1]

$$1 - \frac{2}{3}x - (1 - x)^{\frac{2}{3}} = K_d t$$

$$1 - (1 - x)^{\frac{1}{3}} = K_c t$$

Where:

x : the fraction of metal leached in time.

t : time.

K_d : : diffusion constant.

K_c : chemical constant.

The kinetics of the reactions involved in nickel recovery through H₂SO₄ leaching are crucial for understanding and optimizing the efficiency of the process. The rate of the leaching reactions can be influenced by various parameters, and a fundamental understanding of these kinetics is essential for process control and optimization. Several factors can affect the reaction rate, including temperature, acid concentration, particle size, agitation, and the specific characteristics of the cathode material.

Here's a brief overview of the kinetics and parameters affecting the leaching reactions:

1.initial H₂SO₄ concentration:

The concentration of sulfuric acid influences the reaction rate. It is apparent that the leaching of metals increased gradually with the increase in initial H₂SO₄ concentration. The dissolution of metals follows an upward trend with the rising acid concentration, reaching its peak efficiency at 2M H₂SO₄. However, beyond this concentration, the leaching efficiency of all metals remains relatively constant.

2. Effect of leaching time:

The findings indicate that, within a 60 min approximately 57% of nickel was successfully leached. The highest recovery, reaching around 91.6% was achieved after 120 min. after 120 min, the leaching process reached a point where there was no significant change in recovery, indicating a constant state over time.

3.Effect of pulp density:

The pulp density increases, the leaching efficiency of all metals decreases. The data also imply that the dissolutions of metals remain relatively constant within the pulp density range of 25-100 g/L. the study suggests that a pulp density of 100 g/L is the most suitable value for the remaining experiments.

4.Effect of Temperature:

The leaching efficiency of the metals increased with the increase in temperature. Specifically, at a temperature of 348 K, the recovery percentages for Ni, Co, Fe, Mn, Zn, and REs were found to be 91.6%, 97.8%, 65.5%, 93.5%, 99.2%, and 90.2%, respectively, after leaching the battery powder for 120 minutes.

Understanding the interplay of these factors and their impact on reaction kinetics is essential for designing efficient leaching processes. Experimental studies and mathematical modeling are often employed to determine the optimal conditions for maximizing nickel recovery while minimizing energy consumption and undesirable byproducts.

2.7 The prices of:

Table 5. the price of materials ^[5] ^[12].

Materials	Prices $\frac{USD}{Kg}$
Nickel hydroxide	1500-1800
Nickel sulfate	2500-6000
Sulfuric acid [2M]	25 $[\frac{USD}{L}]$
Nickel	16

2.8 Physical Properties:

Table 6. physical properties:

	Density Kg/L ⁵	State of matter ⁶	Melting temperature (C°) ⁵	boiling temperature (C°) ⁶	Heat capacity J/mol·K ⁷
Nickel	8.9	solid	1455	2740	-
Sulfuric acid	1.8	liquid	10.31	290	-
Cobalt	8.86	solid	1495	2927	-
Cobalt hydroxide	3.597	liquid	168	-	-
Nickel hydroxide	4.1	solid	230	-	-
Nickel sulfate	3.68	solid	-	840	-
Cobalt sulfate	5.45	solid	1195	-	-

2.9 Chemical composition of Spent Ni-MH Battery powder:

Table 7. chemical compositions:

Element	Ni	Co
%(w/w)	58.9	4.6

Table 8: Process efficiency

Yield	91.6% Ni. ¹
selectivity	$K_{1Zn}=0.0065$. ¹ $K_{2Co}=0.0062$. ¹ $K_{3Mn}=0.0051$. ¹ $K_{4Ni}=0.0045$. ¹ $K_{5Fe}=0.0026$. ¹ $K_1>K_2>K_3>K_4>K_5$
Kinetic data	available. ¹
Reaction type	Continuous
Catalyst type	None.
reaction conditions	2 M H ₂ SO ₄ . ¹ 100 g/L pulp density. ¹ 348 K. ¹ 120 min. ¹
The production phase	liquid solid- sulfuric acid solution with solids Slurry. ¹ solid - product
Applicability	This process is innovative, in the article, the process was carried out in samples of hundreds of grams that were laboratory-made It seems that there is factory applicability, but it is necessary to perform a serious upscale, this process is not yet carried out in factory . ¹
Advantages	-Selective Leaching which Targets nickel, cobalt, and copper efficiently.⁴ - Cost Effectiveness and widespread Availability for Sulfuric acid, it is relatively inexpensive and is widely available.⁴ - High Metal Recovery .⁴ - fast kinetics and rapidity of metal extraction.⁴ - High Leaching Efficiency and optimizing resource recovery.⁴ - Economical Usage: Leaches effectively at low concentrations.⁴
Disadvantages	- Environmental Impact: Acidic solutions need careful handling.⁴ - Corrosiveness: Requires specialized equipment due to corrosiveness.⁴ - Byproduct Formation: The process may generate unwanted byproducts.⁴ - Health and Safety Risks: Poses risks, requires strict safety measures.⁴

3. Expanded process Description

3.1 Process break down

The nickel recovery process from spent AA- and AAA-type cylindrical nickel-metal hydride (Ni-MH) batteries involves a systematic approach to electrode material preparation. After discharging the batteries which is last 3 hours, a scutter cutter disassembles them in deionized water, efficiently separating heavier particles, like the steel casing ($\rho = 7.0 - 7.5 \frac{gr}{cm^3}$), from the electrode material ($\rho = 2.5 - 4.0 \frac{gr}{cm^3}$) Subsequent steps include floating away the lighter nylon separator ($\rho_{polypropylene (PP)} = 0.9 \frac{gr}{cm^3}$), wet screening the remaining black powder, and subjecting the electrode powder to a meticulous washing process repeatedly with distilled water at ambient temperature and pressure. This step is crucial for eliminating potential metal contaminants, specifically sodium and potassium.^{2 20 24}

The rigorous washing process prevents these metals from acting as activators in subsequent chemical reactions. After washing, the battery powder undergoes careful drying at $T = 80^\circ C$. Following this, calcination at $T = 550^\circ C$ for 3 hours to remove organic matter, emphasizing a commitment to material purity and quality. After that conducting a cooling process to $T = 75^\circ C$.^{1 2}

The sulfuric acid leaching ($C = 2M$) of metals from the electrode material takes place in a controlled CSTR with external cooling jackets equipped with a magnetic stirrer, sensor-based temperature controller, and a reflux condenser which was fitted to the reactor to avoid the loss due to evaporation. Maintaining stirring conditions minimizes diffusion layers. Continuous in-line monitoring ensures real-time progress, with an automated sampling system handling regular analysis. The leaching reaction ($Ni(OH)_2 + H_2SO_4 \rightarrow NiSO_4 + 2H_2O$) is a cornerstone of the process that occurs at $pH = 1, T = 75^\circ C, P = amb$.^{1 2}

Following leaching, the solution is heated to $T = 90 - 95^\circ C$ in an electric heater to evaporate most of the byproduct water and to increase the concentration of the solutes, next step is cooling at $T = 31.5^\circ C$ in electric cooler for the purpose of $NiSO_4$ crystallization, then the filtration at $T = 30^\circ C$ takes place to separate the precipitated $NiSO_4$ crystal from the remaining solution, the $NiSO_4$ crystal is retained on the filter, while the dissolved $CoSO_4$ and water remain in the filtrate.^{1 2}

The comprehensive process culminates in the production of high-quality nickel, achieved through advanced electrolysis at $E^0 = -0.26 V$ (VS SHE), $T = 30^\circ C$. The electrolysis reaction ($Ni^{2+} + 2e^- \rightarrow Ni$) is pivotal in obtaining this high-quality nickel.^{1 2 20}

3.2 Services & Supplies

Table 9. services and supplies:

Service stream type	supply pressure	supply temperature	purpose	Sign in flow chart
deionized water	1 atm	25[°C]	Separate the electrode material	SS40
Distilled water	1 atm	25[°C]	eliminate any possible metals (e.g., Na and K)	WW80
Hot air	1 atm	90 [°C]	Dry the battery powder from the water	DD110
Cold air	1 atm	5[°C]	Cooling the battery powder	CC160
Oxygen	1 atm	25 [°C]	Used for burning the organic materials	AR135

3.3 Ecology

Table 10: ecology

Flow type	Temperature	Pressure	contain	Treatment
Metallic particles iron case after separation	25 °C	1 atm	Steel, $H_2O_{[l]}$	Recycling iron case to a designated scrap metal recycling facility.
Plastic particles after separation	25 °C	1 atm	Plastic, $H_2O_{[l]}$	Recycling- Production of new plastic products Such as manufacture new battery covers and cases
Water after washing	25 °C	1 atm	$H_2O_{[l]}$ Sodium, potassium	Purification the water from sodium and potassium, and restoration Wastewater treatment system. The sodium and potassium are assembled for factories that use them.

Flow type	Temperature	Pressure	contain	Treatment
Saturated steam after drying	80°C	1 atm	$H_2O_{[g]}$	Relist to the atmosphere
Organic compounds after burning	550°C	1 atm	$H_2O_{[g]}$, $CO_{2[g]}$, $N_{2[g]}$	Relist to the atmosphere
Vapor steam after heating	100°C	1 atm	$H_2O_{[g]}$	Relist to the atmosphere
Filtered solution steam after filtration	31.5°C	1 atm	$CoSO_{4[aq]}$, $H_2O_{[l]}$	Purification the water and restoration Wastewater treatment system. The Co assembled for factories that use it.

3.4 Danger concerns

Table 11: Danger concerns^[22] ^[23]:

material	Autoignition temperature	Flammable limits	Flash point	Health hazard	Flammability hazard	Physical hazard
deionized water	-----	-----	-----	-----	-----	-----
Cold Air	-----	-----	-----	-----	-----	-----
Sulfuric acid [2M]	-----	-----	-----	Highly corrosive to metals	-----	Causes severe skin burns and eye damage.
Hot Air	-----	-----	-----	-----	-----	-----

3.5 Environmental considerations

The ideal place to establish a recycling factory would be in Asia pacific,¹³ as it is a dominant part in the battery production, and that would provide high battery availability-from production scrapes- which reduce the cost of importation, as well as it has a lower energy and labor costs compared to other regions.

It is important that the geographic place of the factory would be near an open sources of water- a shore- due to the large amounts of water required for the process, in addition, it is advisable to place the factory in a roughly far place from the population due to the cursive nature of the sulfuric acid used in the prosses, yet not in a deserted place , a big industrial center would be a good candidate, where there are sewage for waste management, fairly easy workers and personnel accessibility and electricity for large equipment would be available, additional advised location would be near a battery manufactory to reduce the transportation costs.

Overall, the NiMH batteries are considered to be environmentally friendly in comparison to other batteries, and they do not require a special treatment regarding safety and pollution, which opens up the location limitations.

3.6 Flow transport and storage

Table 12: Flow transport and storage

Material type	pipe	container	Work requirements	storage requirements	Sign in flow chart
deionized water	PVC	Stainless steel	$P = 1 \text{ atm}$ $T = 25[^\circ\text{C}]$	$P = \text{amb}$ $T = \text{amb}$	SS40
Distilled water	PVC	Stainless steel	$P = 1 \text{ atm}$ $T = 25[^\circ\text{C}]$	$P = \text{amb}$ $T = \text{amb}$	WW80
Hot air	Stainless steel	Stainless steel	$P = 1 \text{ atm}$ $T = 100[^\circ\text{C}]$	$P = \text{amb}$ $T = 100[^\circ\text{C}]$	DD110
Cold air	PVC	Stainless steel	$P = 1 \text{ atm}$ $T = 10[^\circ\text{C}]$	$P = \text{amb}$ $T = 10[^\circ\text{C}]$	CC160
Sulfuric acid	Stainless steel	Stainless steel	$P = 1 \text{ atm}$ $T = 75[^\circ\text{C}]$	$P = \text{amb}$ $T = 75[^\circ\text{C}]$	LL190
Organic compound	Stainless steel	Stainless steel	$P = 1 \text{ atm}$ $T = 550 - 558[^\circ\text{C}]$	-----	SS50
solution steam	Stainless steel	Stainless steel	$P = 4 \text{ atm}$ $T = 25[^\circ\text{C}]$	-----	SS30
batteries	Stainless steel	Stainless steel	$P = 1 \text{ atm}$ $T = 25[^\circ\text{C}]$	$P = \text{amb}$ $T = \text{amb}$	DI10
the battery powder	Stainless steel	Stainless steel	$P = 1 \text{ atm}$ $T = 80[^\circ\text{C}]$	-----	CC160
leaching reaction product	Stainless steel	Stainless steel	$P = \text{amb}$ $T = 75[^\circ\text{C}]$	-----	LE200
nickel	Stainless steel	Stainless steel	$P = 1 \text{ atm}$ $T = 25[^\circ\text{C}]$	$P = \text{amb}$ $T = 25[^\circ\text{C}]$	EE260

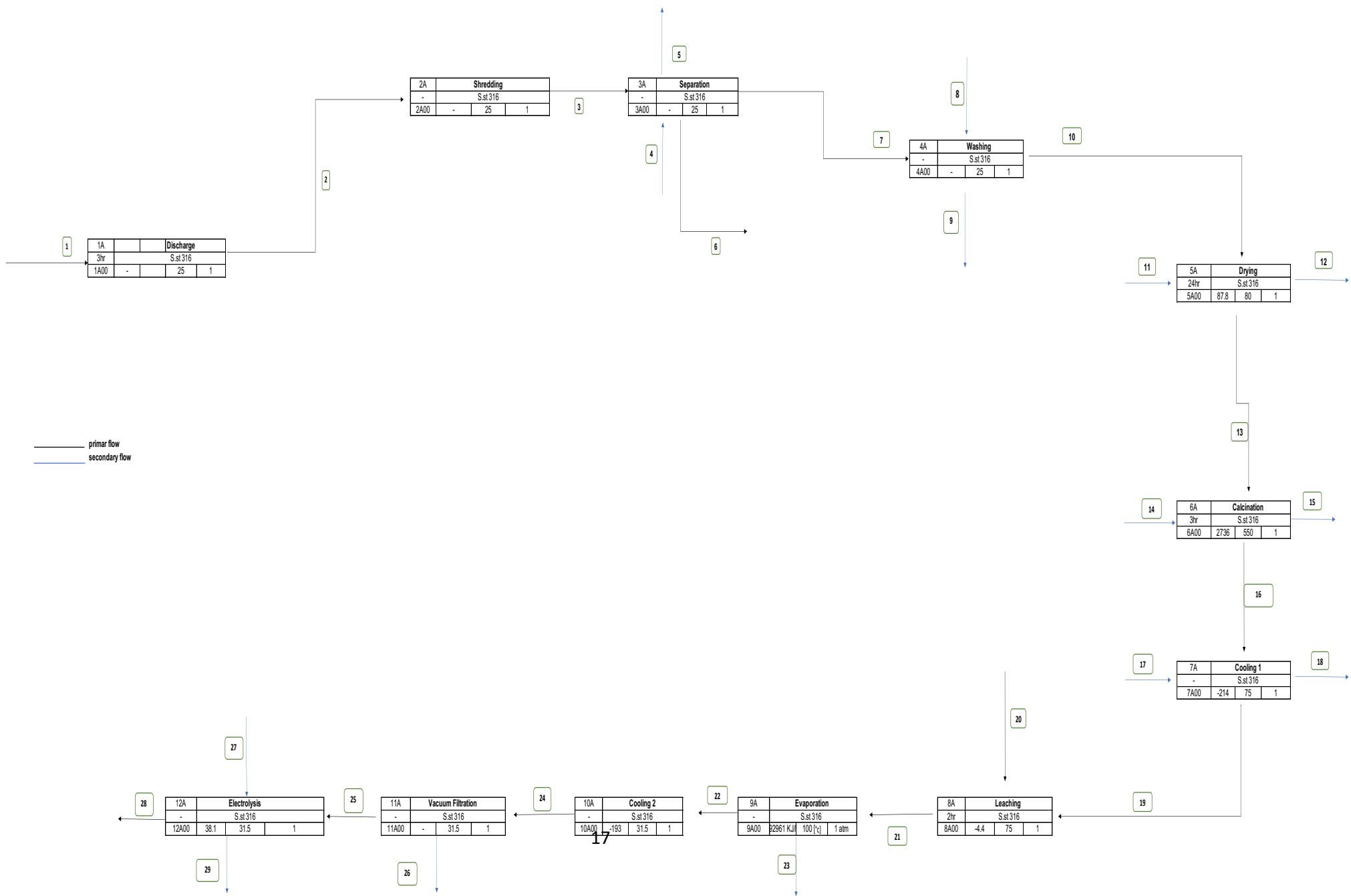
Table 13: Flow transport and storage

Material type	consumed	produced	Storage volume	Sign in flow chart
batteries	$1375 \frac{Kg}{hr}$	-----	1 m ³ container	DI10
Sulfuric acid	$1241.08 \frac{Kg}{hr}$	-----	10 m ³ tank	LL190
Nickel sulfate	$577.5 \frac{Kg}{hr}$	-----	5 m ³	LE240
Cobalt sulfate	$48.125 \frac{Kg}{hr}$	-----	5 m ³	LL250
nickel	-----	$336.06 \frac{Kg}{hr}$	1 m ³ container	EE260

4. Block diagram

Stream No.	Dimensions	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Description		DI10	DS20	SS30	SS40	SS50	SS60	SW70	WW80	WW90	WD100	DD110	DD120	DB130	AR135
Physical state		soild	soild	soild	liquid	liquid+solid	liquid+solid	liquid+solid	liquid	liquid+solid	liquid+solid	gas	gas	solid	gas
Mass flow	Kg/hr	1375	1375	1375	1375	192.5	632.5	1925	1000	1963	962.5	8680	8873	770	72.79
Vol flow	m3/hr	0.3588	0.3588	0.3588	1.3791	0.2030	0.1534	0.9625	1.0030	1.9841	0.3188	8929.12	10317.4	0.2188	16.494
Temperature	C°	25	25	25	25	25	25	25	25	25	25	90	80	80	25
Pressure	atm	1	1	1	1	3	1	1	1	1	5	1	1	1	1
Density	Kg/m3	3833	3833	3833	997	948.5	4123.5	2000.04	997	989.111	3019.175	0.9721	0.86	3518.73	4.413
Viscosity	cP	-	-	-	0.894	0.894	0.894	16.32	0.894	0.877	31.93	-	0.2595	-	0.01837
Heat capacity	KJ/KgK°	-	-	-	4.1816	2.9908	2.3158	2.5015	4.1816	3.943	1.6577	1.011	2.5989	1.031	0.7
Composition															
iron case	%wt	23.00	23.00	23.00	-	-	50.00	-	-	-	-	-	-	-	-
PP	% wt	7.00	7.00	7.00	-	50.00	-	-	-	-	-	-	-	-	-
Nickel hydroxide	% wt	42.00	42.00	42.00	-	-	-	30.00	-	-	60.00	-	-	75.00	-
cobalt hydroxide	% wt	3.50	3.50	3.50	-	-	-	2.50	-	-	5.00	-	-	6.00	-
PEG & PAM	% wt	10.50	10.50	10.50	-	-	-	7.50	-	-	15.00	-	-	19.00	-
K& Na and other	% wt	14.00	14.00	14.00	-	-	-	10.00	-	9.80	-	-	-	-	-
water	% wt	-	-	-	100.00	50.00	50.00	50.00	100.00	90.20	20.00	-	2.20	-	-
sulfuric acid	% wt	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Cold Air	% wt	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Hot Air	% wt	-	-	-	-	-	-	-	-	-	-	100.00	97.80	-	-
N2	% wt	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	% wt	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Nickel sulfate	% wt	-	-	-	-	-	-	-	-	-	-	-	-	-	-
cobalt sulfate	% wt	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Nickel	% wt	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Na2SO4	% wt	-	-	-	-	-	-	-	-	-	-	-	-	-	-
NaCl	% wt	-	-	-	-	-	-	-	-	-	-	-	-	-	-
O2	%wt	-	-	-	-	-	-	-	-	-	-	-	-	-	100

Stream No.	Dimensions	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29
Description		BB140	BC150	CC160	CC170	CL180	LL190	LE200	EC210	CC220	CL230	LE240	LL250	LL260	EE270	EE280
Physical state		gas	solid	gas	gas	solid	liquid	liquid	liquid	gas	liquid+solid	solid	liquid	liquid	solid	liquid
Mass flow	Kg/hr	217.19	625.6	3047	3047	625.6	6269.3	6894.9	1894.9	5000	1894.9	882.55	1012.35	882.55	336.06	1428.94
Vol flow	m3/hr	109.6919	0.1693	2402.9968	3217.5290	0.1540	5.7710	7.2845	2.1704	8361.2040	1.0296	0.4264	0.6173	0.8023	0.0378	1.0576
Temperature	C°	550	550	5	75	75	75	75	100	100	31.5	31.5	31.5	31.5	31.5	31.5
Pressure	atm	1	1	1	1	1	1	3	1	1	1	1	0.08	1	1	1
Density	Kg/m3	1.98	3695.889	1.268	0.947	4061.269	1092.9	946.5115	873.046	0.598	1840.388	2070	1639.92	1100	8900	1351.08
Viscosity	cP	0.03584	0.00235	0.0174	0.02066	-	0.3715	0.3475	0.315	0.2814	12.59	-	7.11	0.32	-	12.56
Heat capacity	KJ/KgK°	2.5256	0.7214	1.005	1.009	0.7214	3.6899	3.44	1.4839	4.2053	1.4884	0.591	2.7675	0.849	0.46	0.42
Composition																
iron case	%wt	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
PP	% wt	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Nickel hydroxide	% wt	-	92.30	-	-	92.30	-	0.70	2.55	-	2.55	-	4.77	-	-	-
cobalt hydroxide	% wt	-	7.70	-	-	7.70	-	-	-	-	-	-	-	-	-	-
PEG & PAM	% wt	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
K& Na and other	% wt	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
water	% wt	25.00	-	-	-	-	82.00	77.75	19.05	100.00	19.05	-	35.64	85.00	-	35.00
sulfuric acid	% wt	-	-	-	-	-	18.00	7.55	27.50	-	27.50	-	51.42	-	-	-
Cold Air	% wt	-	-	100.00	-	-	-	-	-	-	-	-	-	-	-	-
Hot Air	% wt	-	-	-	100.00	-	-	-	-	-	-	-	-	-	-	-
N2	% wt	5.50	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	% wt	69.50	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Nickel sulfate	% wt	-	-	-	-	-	-	12.80	46.50	-	46.50	100.00	-	-	-	-
cobalt sulfate	% wt	-	-	-	-	-	-	1.20	4.40	-	4.40	-	8.17	-	-	-
Nickel	% wt	-	-	-	-	-	-	-	-	-	-	-	-	-	100.00	-
Na2SO4	% wt	-	-	-	-	-	-	-	-	-	-	-	-	-	-	60.00
NaCl	% wt	-	-	-	-	-	-	-	-	-	-	-	-	15	-	5
O2	%wt	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-



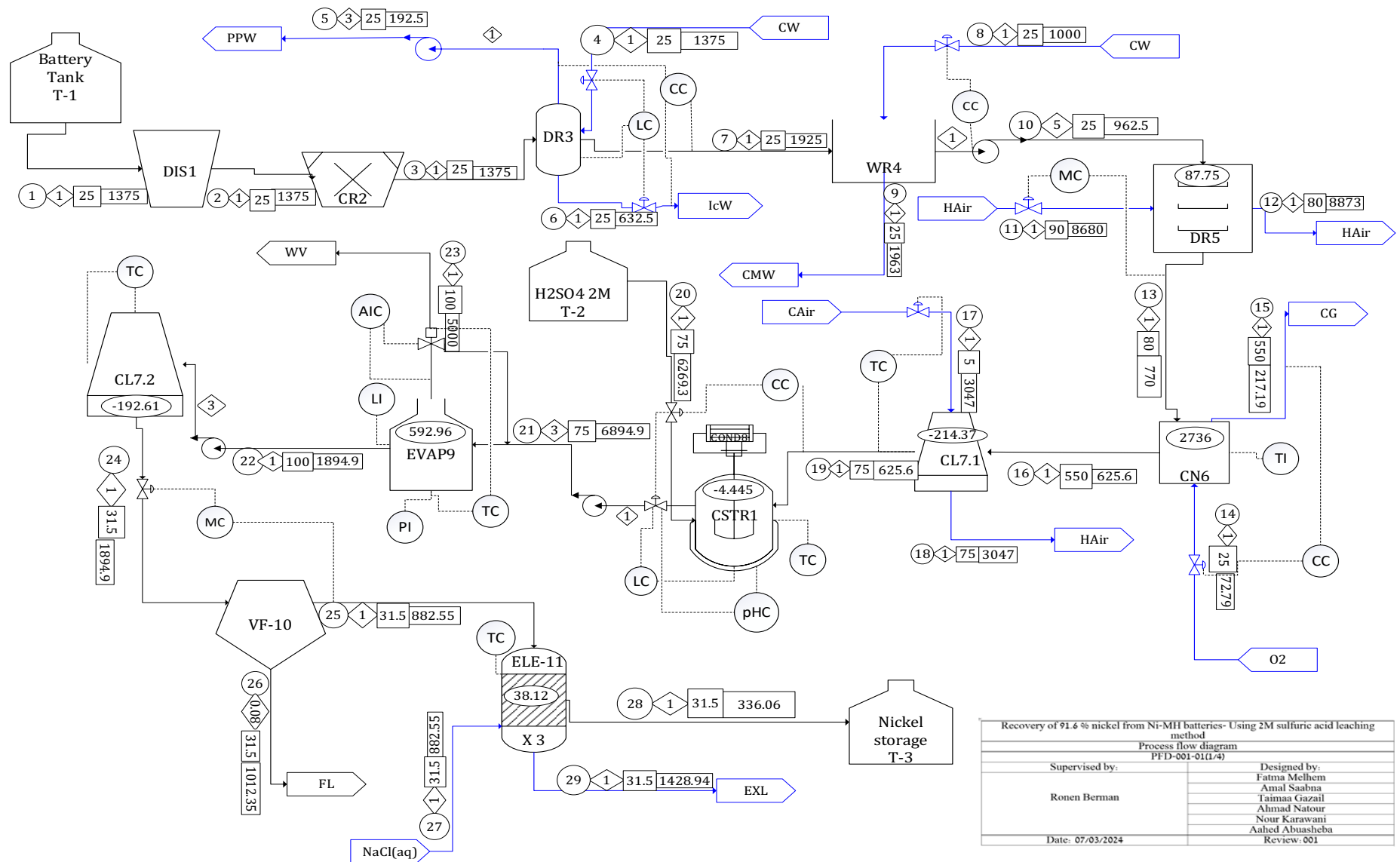
Legend

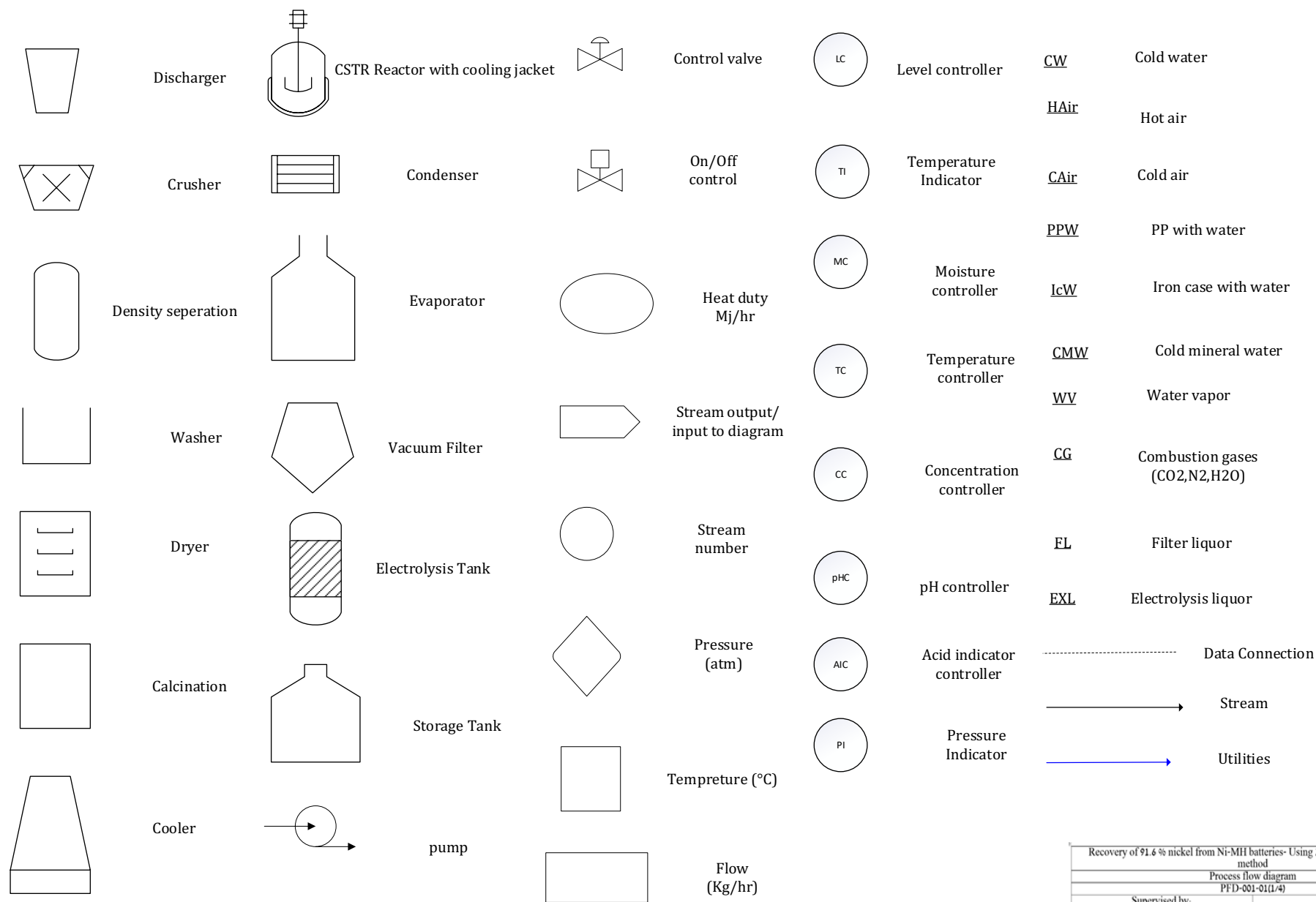
stage	Description
Time	MOC (material of constructions)
Eq. Number	Energy (MJ/hr) T (°C) P (atm)

— primary flow
— secondary flow

4		Recovery of 91.6 % nickel from Ni-MH batteries- Using 2M sulfuric acid leaching method -BLOCK DIAGRAM	
No.	7.2.24	Description	
Project 5			
Nickel recovery from Ni-MH batteries using sulfuric acid leaching		Drawn and Design:	Fatma Melhem, Aml Saabna, Taimaa Gzail, Aahed Abuasheba, Nour Karawani, Ahmad Natour
BLOCK DIAGRAM		Checked:	10

5. PFD





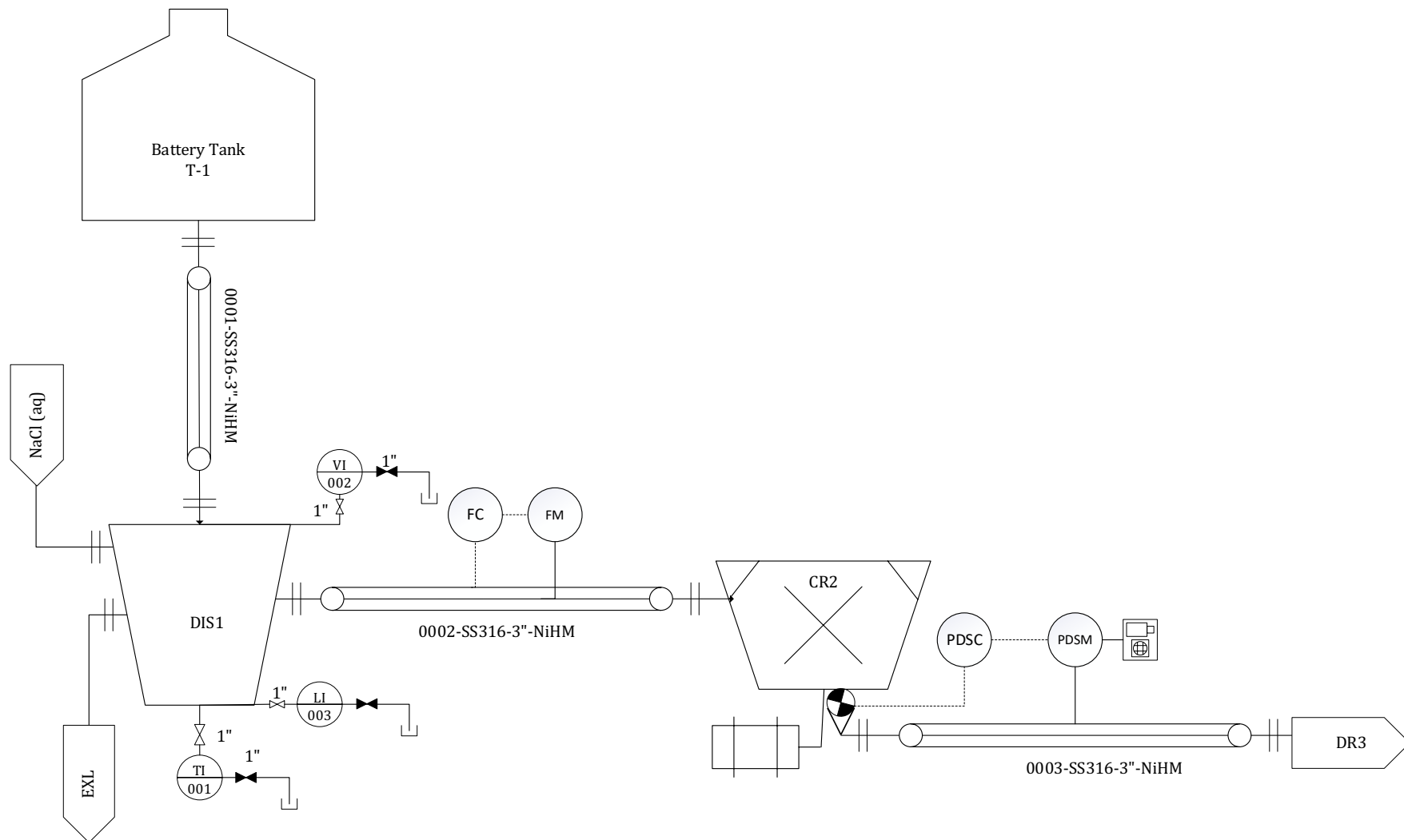
Recovery of 91.6 % nickel from Ni-MH batteries- Using 2M sulfuric acid leaching method	
Process flow diagram	
PFD-001-01(1/4)	
Supervised by:	Designed by:
Ronen Berman	Fatma Melhem
	Amal Saabna
	Taimaa Gazail
	Ahmad Natour
	Nour Karawani
Aahed Abuasheba	
Date: 07/03/2024	Review: 001

Name	Battery Tank	Discharger	Crusher	Density separation	Washer	Dryer	Calcination	Cooler 1	CSTR Reactor with cooling jacket
Symbol	T-1	DIS1	CR2	DR3	WR4	DR5	CN6	CL7.1	CSTR1
Function	For Ni-MH battery storage	Discharging Ni-MH Battery	Shredding the battery	Separation the battery compounds	Washing to eliminating Na, K	Drying the battery powder	Purification by burning the organic matters	Cooling the black powder	Leaching process
Construction material	S.S	S.S	S.S	S.S	S.S	S.S	S.S	S.S	S.S
Chemical	Ni-MH Battery	Ni-MH Battery	Ni-MH Battery	Ni-MH Battery compounds and water	Ni-MH Battery compounds and water	Ni-MH Battery compounds and water	Ni-MH Battery compounds	Ni-MH Battery black powder	Black powder and Sulfuric acid
Typical size	1 m ³	2 m ³	2 m ³	5 m ³	2 m ³	70m ³	60 m ³	50 m ³	20 m ³

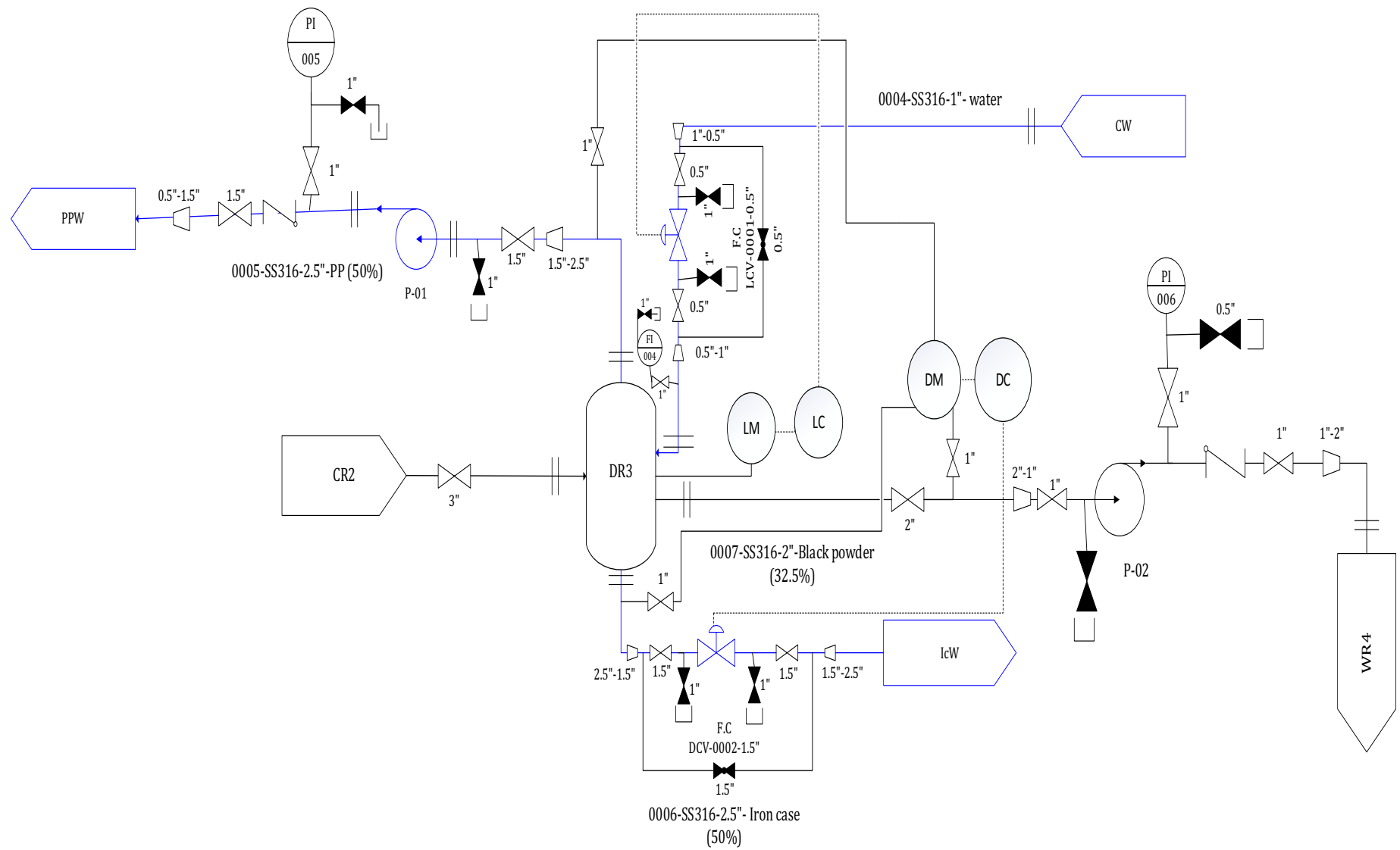
Name	Condenser	H2SO4 Tank	Evaporator	Cooler 2	Vacuum Filter	Electrolysis Tank	Nickel storage
Symbol	COND8	T-2	EVAP9	CL7.2	VF-10	ELE-11	T-3
Function	To Re-condense the solution vapor	Sulfuric acid storage	To evaporate water from the solution	Crystalizing nickel sulfate	To separate the crystal from the solution	To extract Nickel metal	For nickel storage
Construction material	S.S	S.S	S.S	S.S	S.S	S.S	S.S
Chemical	Sulfuric acid	Sulfuric acid	Cobalt sulfate, nickel sulfate and H2SO4 2M	Liquor and nickel sulfate	liquor and nickel sulfate crystal	nickel sulfate crystal and NaCl solution	Pure nickel metal
Typical size	11 m ³	10 m ³	30 m ³	5 m ³	5 m ³	4 m ³	1 m ³

Recovery of 91.6 % nickel from Ni-MH batteries- Using 2M sulfuric acid leaching method	
Process flow diagram	
PFD-001-01(1.4)	
Supervised by:	Designed by:
Ronen Berman	Fatma Melhem
	Amal Sanbna
	Taimaa Gazal
	Ahmad Natour
	Nour Karawani
Aahed Abuasheba	
Date: 07/03/2024	Review: 001

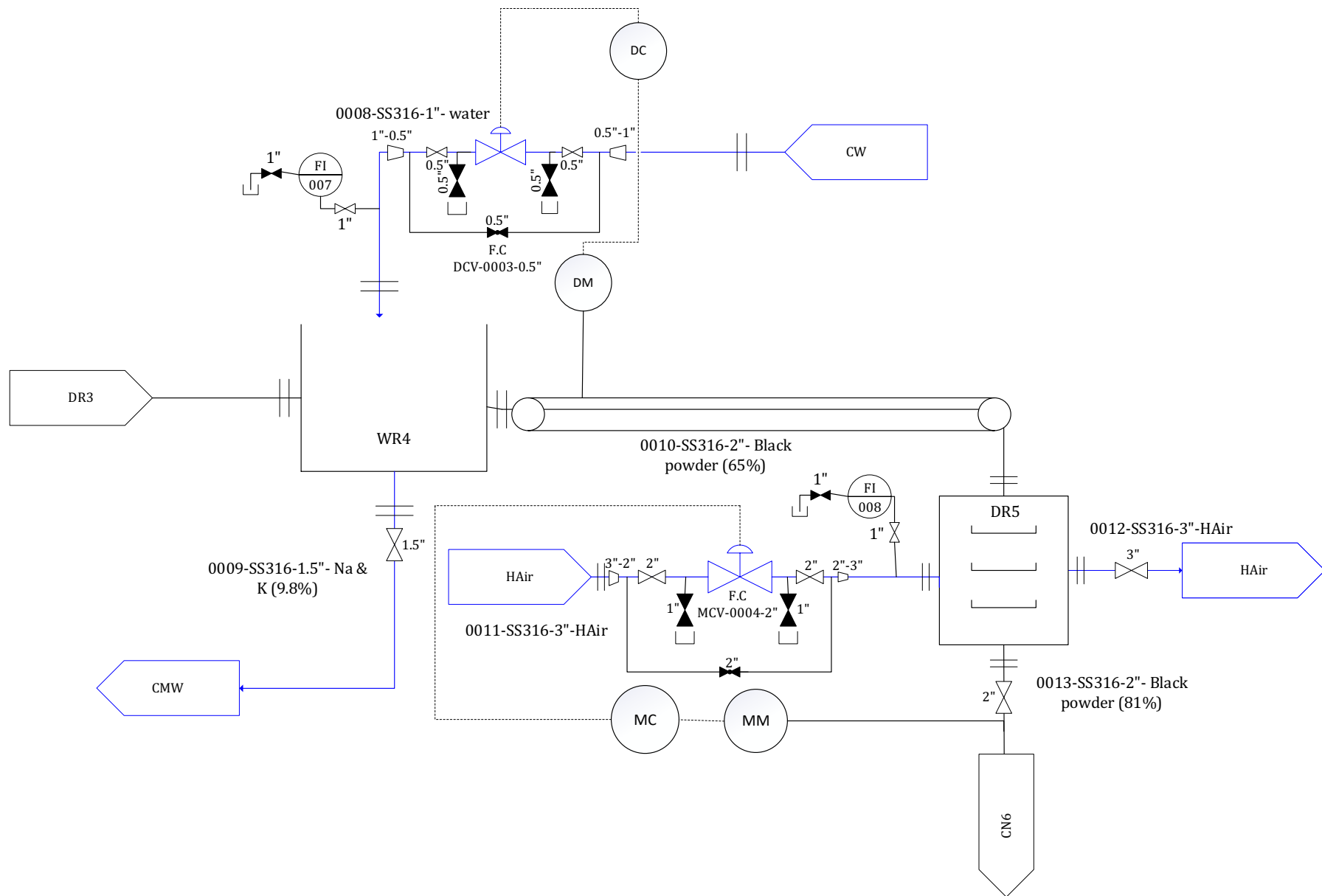
6. P&ID



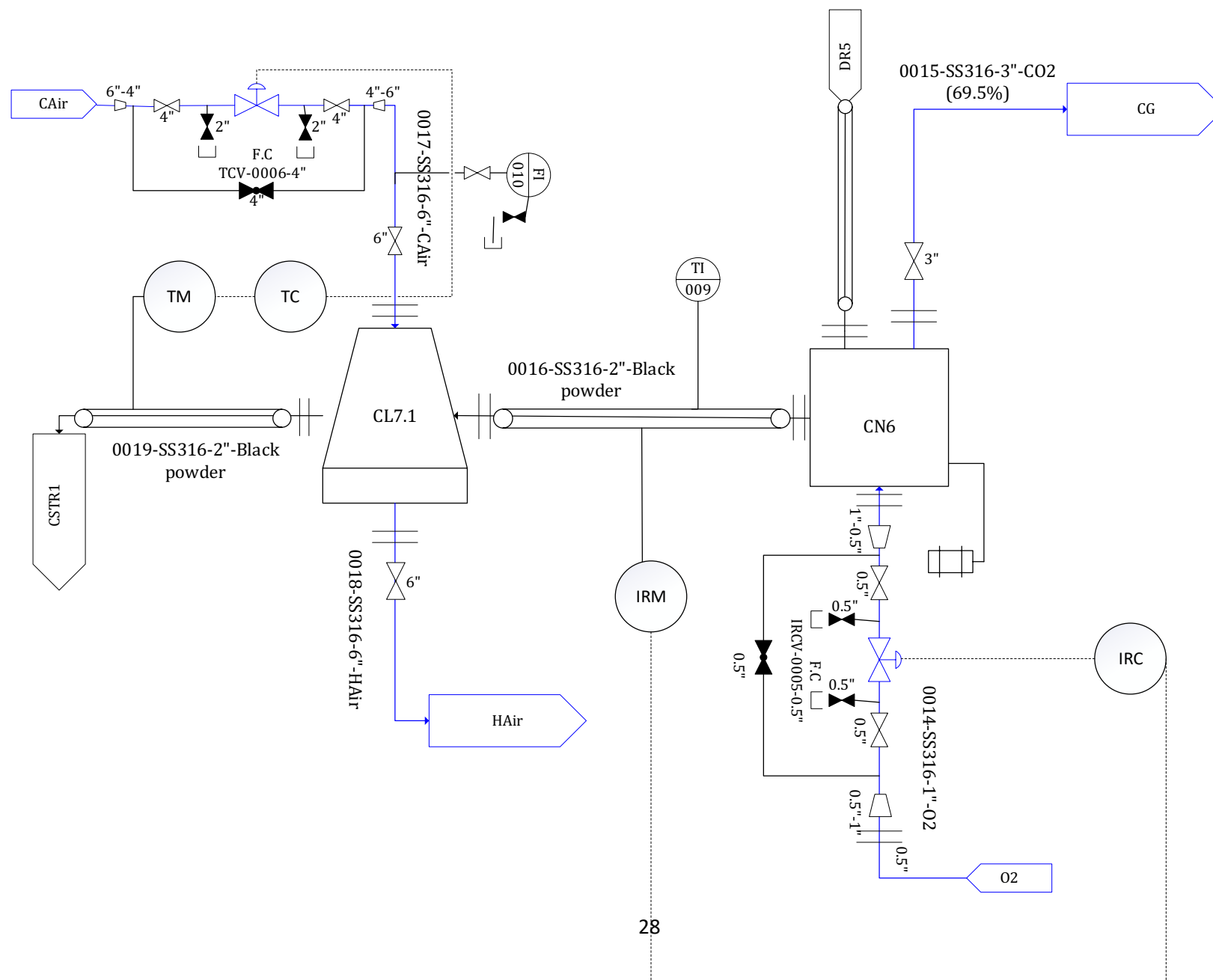
Facility name: Nickel recovery from Ni-MH batteries using H ₂ SO ₄ leaching.	
Sketch name: P&ID	
Sketch number: 1/8 P&ID	
Supervisor: Mr. Ronen Berman	Engineers name: Aml Saabna, Ahmad Natour, Taimaa Gzail, Aahed Abuasheba, Fatma Melhem & Nour Karawani.
Date: 9.6.24	Edition: 010



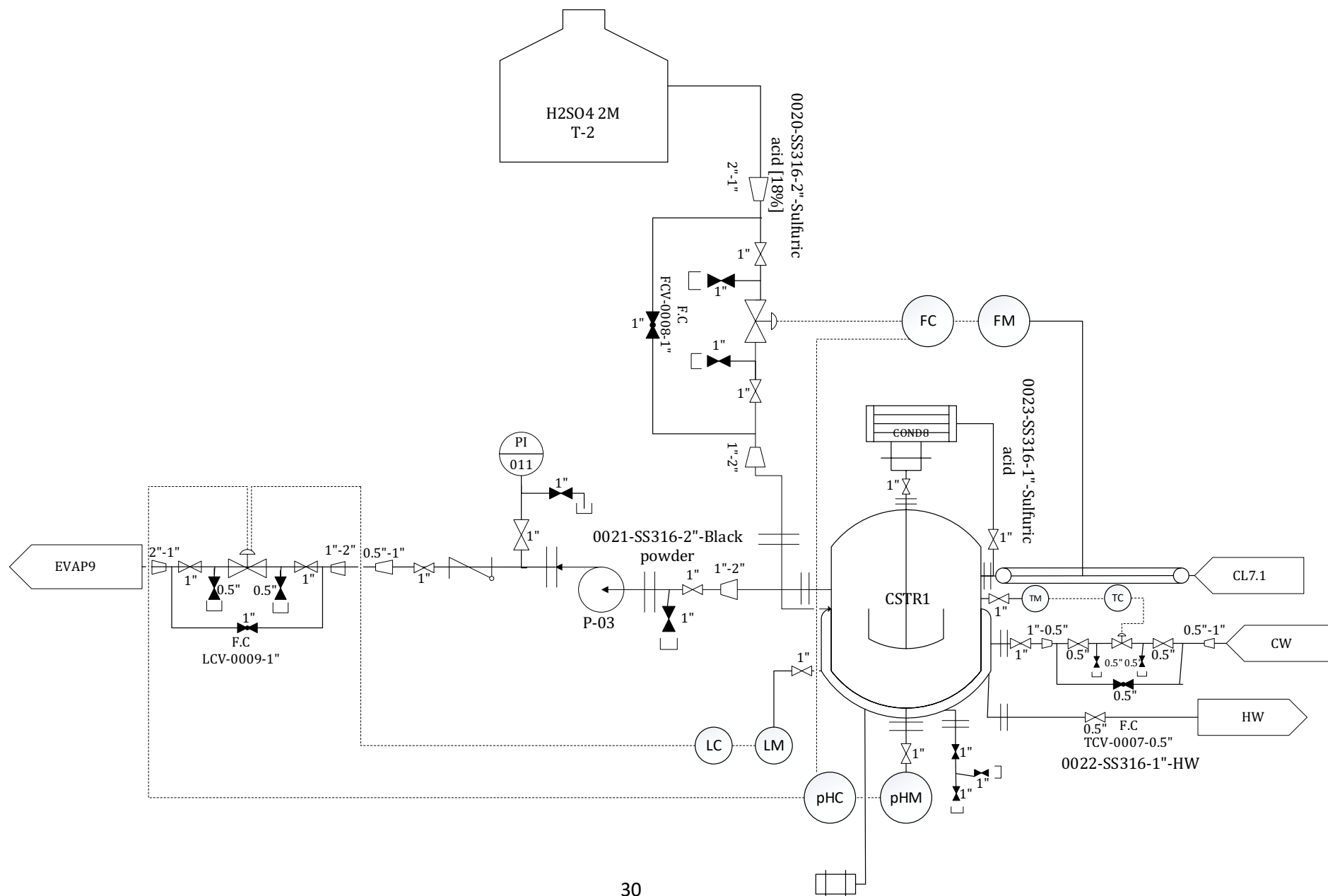
Facility name: Nickel recovery from Ni-MH batteries using H ₂ SO ₄ leaching.	
Sketch name: P&ID	
Sketch number: 2/8 P&ID	
Supervisor: Mr. Ronen Berman	Engineers name: Aml Saabna, Ahmad Natour, Taimaa Gzail, Aahed Abuasheba, Fatma Melhem & Nour Karawani.
Date: 9.6.24	Edition: 010



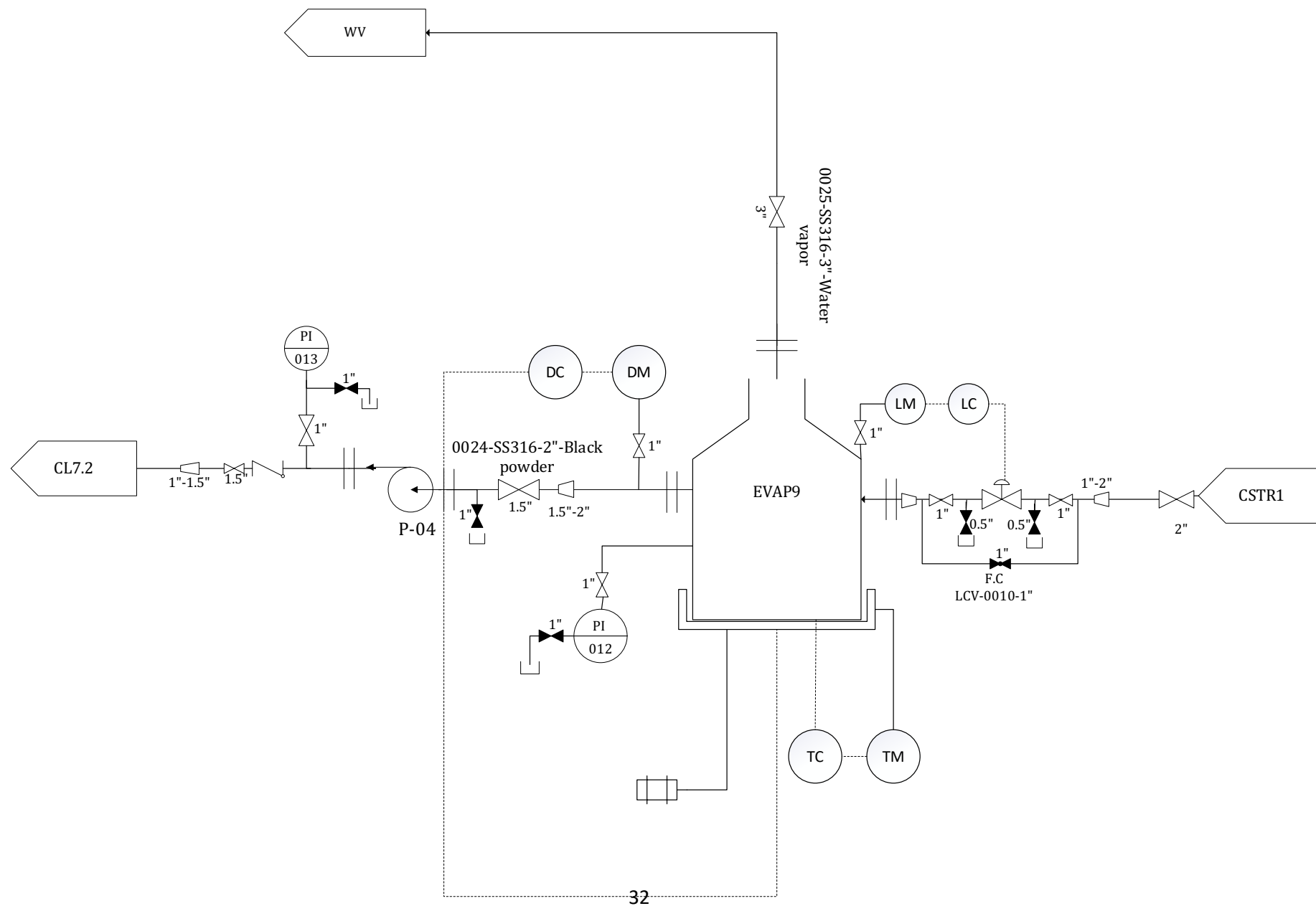
Facility name: Nickel recovery from Ni-MH batteries using H2SO4 leaching.	
Sketch name: P&ID	
Sketch number: 3/8 P&ID	
Supervisor: Mr. Ronen Berman	Engineers name: Aml Saabna, Ahmad Natour, Taimaa Gzail, Aahed Abuasheba, Fatma Melhem & Nour Karawani.
Date: 9.6.24	Edition: 010



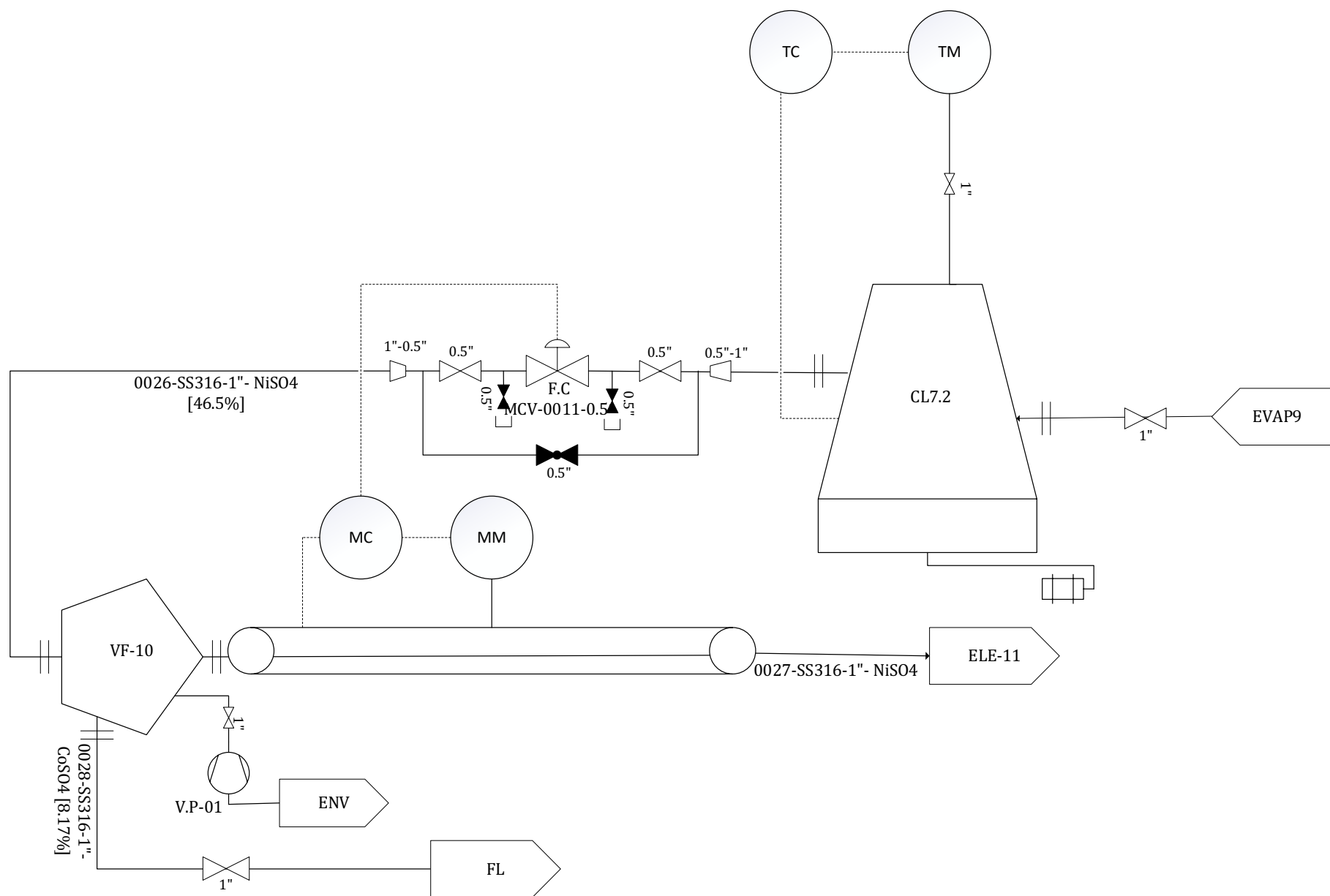
Facility name: Nickel recovery from Ni-MH batteries using H ₂ SO ₄ leaching.	
Sketch name: P&ID	
Sketch number: 4/8 P&ID	
Supervisor: Mr. Ronen Berman	Engineers name: Aml Saabna, Ahmad Natour, Taimaa Gzail, Aahed Abuasheba, Fatma Melhem & Nour Karawani.
Date: 9.6.24	Edition: 010



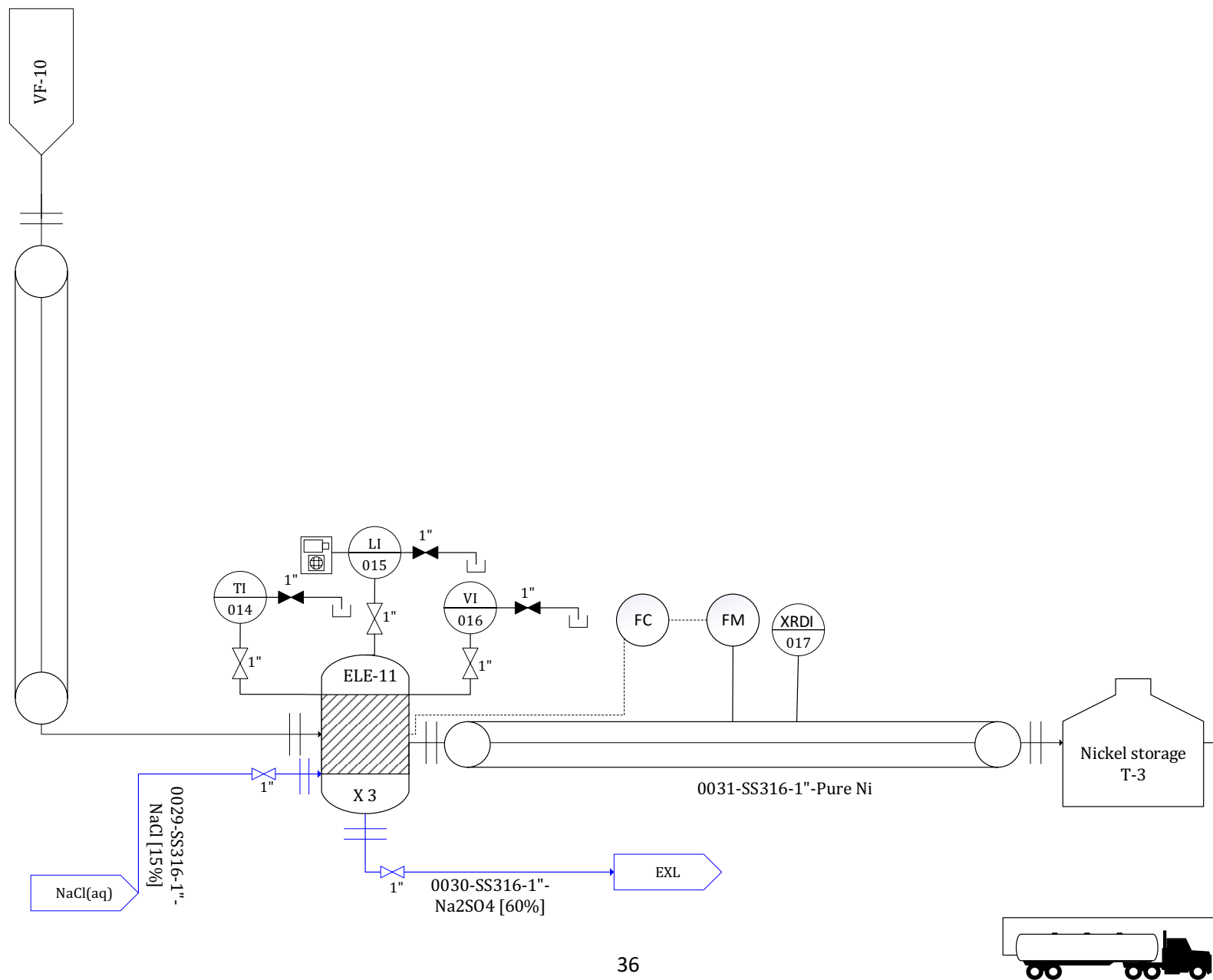
Facility name: Nickel recovery from Ni-MH batteries using H ₂ SO ₄ leaching.	
Sketch name: P&ID	
Sketch number: 5/8 P&ID	
Supervisor: Mr. Ronen Berman	Engineers name: Aml Saabna, Ahmad Natour, Taimaa Gzail, Aahed Abuasheba, Fatma Melhem & Nour Karawani.
Date: 9.6.24	Edition: 010



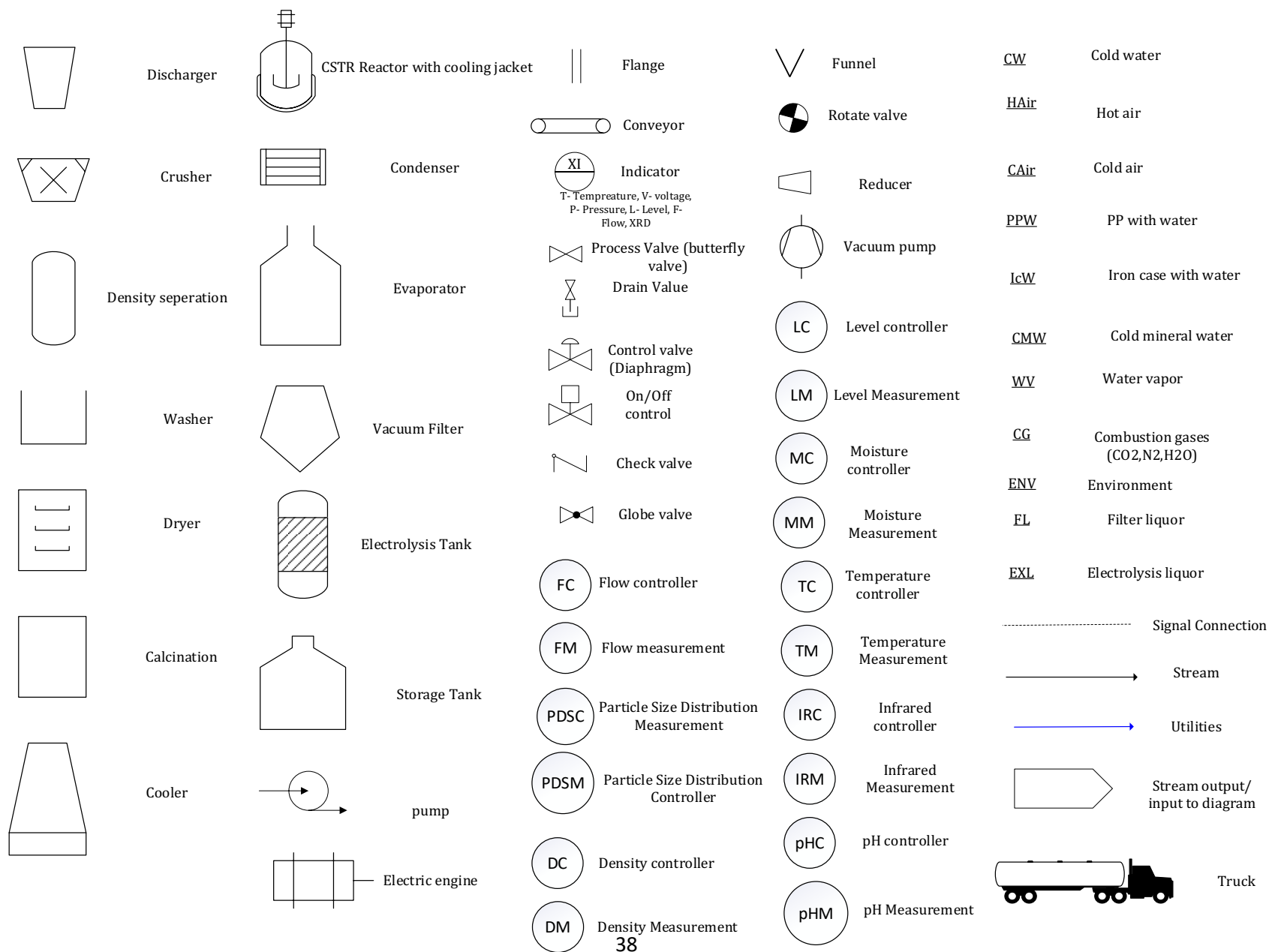
Facility name: Nickel recovery from Ni-MH batteries using H2SO4 leaching.	
Sketch name: P&ID	
Sketch number: 6/8 P&ID	
Supervisor: Mr. Ronen Berman	Engineers name: Aml Saabna, Ahmad Natour, Taimaa Gzail, Aahed Abuasheba, Fatma Melhem & Nour Karawani.
Date: 9.6.24	Edition: 010



Facility name: Nickel recovery from Ni-MH batteries using H ₂ SO ₄ leaching.	
Sketch name: P&ID	
Sketch number: 7/8 P&ID	
Supervisor: Mr. Ronen Berman	Engineers name: Aml Saabna, Ahmad Natour, Taimaa Gzail, Aahed Abuasheba, Fatma Melhem & Nour Karawani.
Date: 9.6.24	Edition: 010



Facility name: Nickel recovery from Ni-MH batteries using H2SO4 leaching.	
Sketch name: P&ID	
Sketch number: 8/8 P&ID	
Supervisor: Mr. Ronen Berman	Engineers name: Aml Saabna, Ahmad Natour, Taimaa Gzail, Aahed Abuasheba, Fatma Melhem & Nour Karawani.
Date: 9.6.24	Edition: 010



7.Equipment Specification

				TANK/VESSEL				Tag.No.:			
				PERFORMANCE DATA				T - 01			
				PROJECT NAME: Nickel recovery from Ni-MH batteries using H2SO4 leaching.				SHEET		1 of 2	
				PROJECT No.: 1				REV.		1	
				PLANT: 1				BY:		Team 5	
								DATE:		09/06/2024	
NAME: Battery Tank											
FUNCTION: For Ni-MH battery storage											
OPERATION: Continuous											
MATERIALS											
PROCESS: Storing batteries and transferring them to the next phase, which is discharge											
UTILITIES: XX											
PERFORMANCE INFORMATION											
			UNITS		NORMAL OPERATION			DESIGN BASIS			
WORKING VOLUME			m ³		0.359			0.5			
FREEBOARD			m ³		0.641			0.5			
TOTAL VOLUME			m ³		1			1			
INTERNAL PRESSURE			atm		0.3-1			0-1			
INTERNAL TEMPERATURE			°C		25			25			
DENSITY			Kg/m ³		3600-4000			4000			
EQUIPMENT INFORMATION											
TYPE: Vertical					FLANGES:			ANSI #150			
					GASKETS:			EPDM			
DIMENSIONS: Diam: 1 m					HEAT/COOL TYPE:			No			
High/Long: 1.28 m					AGITATOR:			No			
					INSTRUMENT CONN.:			Yes			
					VENT,PRESS.RELIEF:			Yes*			
					CODE:						
					INSULATION:			No			
					OTHERS:						
Rev.	Description			Proc.Eng.	Date	Proc.Man.	Date	Proc.Eng.	Date	DSBG Appr.	Date
1	T-01			Team 4	09/06/24						

								Tag.No.:	
				TANK/VESSEL				DIS1	
				PERFORMANCE DATA					
								SHEET	1 of 2
				PROJECT NAME: Nickel recovery from Ni-MH batteries using H2SO4 leaching.				REV.	1
				PROJECT No.: 1				BY:	Team 5
				PLANT: 1				DATE:	09/06/2024
NAME:		Discharge							
FUNCTION:		Discharging Ni-MH Battery							
OPERATION:		Continuous							
MATERIALS									
PROCESS:		The discharge process involves the oxidation of the anode (metal hydride) and the reduction of the cathode (nickel oxyhydroxide) through the movement of electrons and ions via the electrolyte. This process generates an electric current that allow the stored energy in the battery to be used.							
UTILITIES:		XXX							
PERFORMANCE INFORMATION									
		UNITS		NORMAL OPERATION				DESIGN BASIS	
WORKING VOLUME		m^3		0.359				0.4	
FREEBOARD		m^3		0.641				0.6	
TOTAL VOLUME		m^3		1				1	
INTERNAL PRESSURE		atm		0.3-1				0-1	
INTERNAL TEMPERATURE		°C		25				25	
DENSITY		Kg/m³		3600-4000				4000	
EQUIPMENT INFORMATION									
TYPE:		Vertical		FLANGES:		ANSI #150			
				GASKETS:		EPDM			
DIMENSIONS:		Diam:	1	m	HEAT/COOL TYPE:		No		
		High/Long:	1.28	m	AGITATOR:		No		
				INSTRUMENT CONN.:		Yes			
				VENT,PRESS.RELIEF:		Yes*			
				CODE:					
				INSULATION:		No			
				OTHERS:					
Rev.	Description	Proc.Eng.	Date	Proc.Man.	Date	Proc.Eng.	Date	DSBG Appr.	Date
1	DIS1	Team 5	09/06/24						

				TANK/VESSEL PERFORMANCE DATA				Tag.No.: DIS1	
PROJECT NAME: Nickel recovery from Ni-MH batteries using H2SO4 leaching.				SHEET 2 of 2				REV. 1	
PROJECT No.: 1				BY: Team 5				DATE: 09/06/2024	
PLANT: 1									

Diagram of a trapezoidal tank labeled DIS1. The tank has a bottom width of 1 m and a height of 1.28 m. An inlet pipe is at the top center, and an outlet pipe is on the right side. The tank is labeled DIS1 inside.

MARK	DIAM.	SERVICE	FLG.#	REMARKS
A	3"	battereis inlet to the discharge	ANSI #150	
B	3"	battereis outlet from the discharge	ANSI #150	
C	1"	Temperature control	ANSI #150	
D	1"	Voltage control	ANSI #150	

Rev.	Description	Proc.Eng.	Date	Proc.Man.	Date	Proc.Eng.	Date	DSBG Appr.	Date
1	DIS1	Team 5	03/06/24						

								Tag.No.:		
				TANK/VESSEL				CR-2		
				PERFORMANCE DATA						
								SHEET	1 of 2	
				PROJECT NAME: Nickel recovery from Ni-MH batteries usi				REV.	1	
				PROJECT No.: 5				BY:	Team 5	
				PLANT: 1				DATE:	9/6/24	
NAME:		shreding								
FUNCTION:		Cutting the batteries into different size								
OPERATION:		Continuous								
MATERIALS										
PROCESS:		After the disassembly stage, the batteries are transferred via a conveyor to the shredding machine, where they are fed into the shredder through a dedicated inlet. Inside the machine, sharp rotating blades or hammer-like mechanisms create high mechanical pressure that cuts or crushes the battery into smaller pieces.								
UTILITIES:		xxx								
PERFORMANCE INFORMATION										
		UNITS	NORMAL OPERATION				DESIGN BASIS			
WORKING VOLUME		lit	1000				1100			
FREEBOARD		lit	1000				900			
TOTAL VOLUME		lit	2000				2000			
INTERNAL PRESSURE		barg	0.2-1				0-1.7			
INTERNAL TEMPERATURE		°C	25				25			
DENSITY		Kg/m ³	3833				3833			
EQUIPMENT INFORMATION										
TYPE:	Horizontal				FLANGES:		ANSI #150			
					GASKETS:		EDPM			
DIMENSIONS:	Diam:	800	mm	HEAT/COOL TYPE:		No				
	High/Long:	1600	mm	AGITATOR:		No				
MAT'L OF CONSTRUCTION				INSTRUMENT CONN.:		Yes				
HEADS:	SS-316L	CorrAllow:	1mm	VENT,PRESS.RELIEF:		Yes*				
SHELL:	Aluminum	CorrAllow:	1mm	CODE:						
SUPPORT:	Steel	CorrAllow:	1mm	INSULATION:		No				
JACKET:		CorrAllow:	1mm	OTHERS:						
Rev.	Description		Proc.Eng.	Date	Proc.Man.	Date	Proc.Eng.	Date	DSBG Appr.	Date
1	CR-2		Team 5	9/6/24						

[illegible]

								Tag.No.:	
				TANK/VESSEL				DR3	
				PERFORMANCE DATA					
								SHEET	1 of 2
				PROJECT NAME: Nickel recovery from Ni-MH batteries usi				REV.	1
				PROJECT No.:				BY:	Team 5
				PLANT:				DATE:	9/6/24
NAME:		Density seperation							
FUNCTION:		Separates the different components of the battery based on their density differences.							
OPERATION:		Continuous							
MATERIALS									
PROCESS:		The iron case and PP are separated from the other components of the battery using a density-based separator. Due to the differences in the densities of the separated materials, the iron case will settle at the bottom of the separator because of its higher density, while the PP will float to the top due to its lower density. The remaining components will distribute themselves in the middle of the separator based on their respective densities.							
PERFORMANCE INFORMATION									
		UNITS	NORMAL OPERATION				DESIGN BASIS		
WORKING VOLUME		lit	4121				4200		
FREEBOARD		lit	879				800		
TOTAL VOLUME		lit	5000				5000		
INTERNAL PRESSURE		barg	0.2-1				0-1.7		
INTERNAL TEMPERATURE		°C	25				25		
DENSITY		Kg/m ³	900-1000				800-1100		
EQUIPMENT INFORMATION									
TYPE:		Vertical				FLANGES:		ANSI #150	
						GASKETS:		EDPM	
DIMENSIONS:		Diam:	1000	mm		HEAT/COOL TYPE:		No	
		High/Long:	6400	mm		AGITATOR:		No	
MAT'L OF CONSTRUCTION						INSTRUMENT CONN.:		Yes	
HEADS:		SS-316L		CorrAllow:	1mm	VENT,PRESS.RELIEF:		NO*	
SHELL:		SS-316L		CorrAllow:	1mm	CODE:			
SUPPORT:		SS-316L		CorrAllow:	1mm	INSULATION:		No	
JACKET:				CorrAllow:	1mm	OTHERS:			
Rev.	Description			Proc.Eng.	Date	Proc.Man.	Date	Proc.Eng.	Date
1	DR3			Team 5	9/6/24				

				TANK/VESSEL				Tag.No.:			
				PERFORMANCE DATA				DR3			
PROJECT NAME:				Nickel recovery from Ni-MH batteries usi				SHEET		2 of 2	
PROJECT No.:				5				REV.		1	
PLANT:				1				BY:		Team 5	
								DATE:		9/6/24	

								Tag.No.:	
				washer				WR-4	
				PERFORMANCE DATA					
				PROJECT NAME: Nickel recovery from NI-MH batteries by H2SO4 leaching				SHEET	1 of 2
				PROJECT No.: 1				REV.	2
				PLANT: 1				BY:	Team 5
								DATE:	9/6/24
NAME: washing									
FUNCTION:									
OPERATION: Continuous									
MATERIALS									
PROCESS: wash out Na and K ions and filter black powder									
UTILITIES:									
PERFORMANCE INFORMATION									
		UNITS		NORMAL OPERATION			DESIGN BASIS		
WORKING VOLUME		lit		1500			1800		
FREEBOARD		lit		500			200		
TOTAL VOLUME		lit		2000			2000		
INTERNAL PRESSURE		barg		0.05-1			0-1		
INTERNAL TEMPERATURE		°C		20			50		
DENSITY		Kg/m ³		4130			4200		
EQUIPMENT INFORMATION									
TYPE:		Horizontal							
DIMENSIONS:		Diam: 2000 mm		FLANGES:			ANSI#150		
		High/Long: 550 mm		GASKETS:			EPDM		
funnel		Diam: 2000 mm		HEAT/COOL TYPE:			No		
		High/Long: 300 mm		AGITATOR:			No		
MAT'L OF CONSTRUCTION					INSTRUMENT CONN.:			Yes	
HEADS:		SS-316L		CorrAllow	1mm		VENT,PRESS.RELIEF: Yes		
SHELL:		SS-316L		CorrAllow	1mm		CODE:		
SUPPORT:		*		CorrAllow	1mm		INSULATION: No		
JACKET:				CorrAllow	1mm		OTHERS:		
Rev.	Description	Proc.Eng.	Date	Proc.Man.	Date	Proc.Eng.	Date	DSBG Appr.	Date
1	Washer WR-4	Team 5	9/6/24						

					washer			Tag.No.:		
					PERFORMANCE DATA			WR-4		
					PROJECT NAME:			Nickel recovery from NI-MH batteries by H2SC	SHEET	2 of 2
					PROJECT No.:			1	REV.	2
					PLANT:			1	BY:	Team 5
									DATE:	9/6/24
MARK	DIAM.	SERVICE				FLG.#		REMARKS		
1	2"	black powder inlet				ANSI #150				
2	2"	black powder outlet				ANSI #150				
3	1"	water inlet				ANSI #150				
4	1.5"	Na & K with water outlet				ANSI #150				
Rev.	Description		Proc.Eng.	Date	Proc.Man.	Date	Proc.Eng.	Date	DSBG Appr.	Date
2	Washer WR-4		Team 5	9/6/24						

										Tag.No.:	
Calcination										CN6	
PERFORMANCE DATA											
										SHEET	1 of 2
PROJECT NAME: Nickel recovery from NI-MH batteries by H2SC										REV.	2
PROJECT No.: 1										BY:	Team 5
PLANT: 1										DATE:	9/6/24
NAME:		Calcination								MFS	
FUNCTION:		Extraction of Organic materials									
OPERATION:		Continuous									
MATERIAL HANDLED:		xxx									
PERFORMANCE INFORMATION											
		UNITS	NORMAL OPERATION				DESIGN BASIS				
WORKING VOLUME		lit	52000				55000				
FREEBOARD		lit	8000				5000				
TOTAL VOLUME		lit	60000				60000				
INTERNAL PRESSURE		barg	0.05-1				0-1				
INTERNAL TEMPERATURE		°C	550				700				
DENSITY		Kg/m ³	3500				3600				
HEAT		Mj/kg	2736								
EQUIPMENT INFORMATION											
TYPE:		Horizontal									
						FLANGES:		ANSI#150			
DIMENSIONS:		Long:	5000	mm		GASKETS:		EPDM			
		High:	4000	mm		HEAT/COOL TYPE:		No			
		width:	3000	mm		AGITATOR:		No			
MAT'L OF CONSTRUCTION						INSTRUMENT CONN.:		Yes			
HEADS:		SS-316L	CorrAllow	1mm		VENT,PRESS.RELIEF:		Yes			
SHELL:		SS-316L	CorrAllow	1mm		CODE:					
SUPPORT:		*	CorrAllow	1mm		INSULATION:		No			
JACKET:			CorrAllow	1mm		OTHERS:					
Rev.	Description		Proc.Eng.	Date	Proc.Man.	Date	Proc.Eng.	Date	DSBG Appr.	Date	
1	Calcination CN-6		Team 5	9/6/24							

Calcination							Tag.No.:		
PERFORMANCE DATA							CN6		
PROJECT NAME:				Nickel recovery from Ni-MH batteries by H2SC			SHEET		1 of 2
PROJECT No.:				1			REV.		2
PLANT:				1			BY:		Team 5
							DATE:		9/6/24

MARK	DIAML	SERVICE	FLG.#	REMARKS
1	2"	black powder inlet	ANSI #150	
2	1"	O2 inlet	ANSI #150	
3	3"	gas outlet	ANSI #150	
4	3"	black powder outlet	ANSI #150	

Rev.	Description	Proc.Eng.	Date	Proc.Man.	Date	Proc.Eng.	Date	DSBG Appr.	Date
1	Calcination CN6	Team 5	9/6/24						

				T&S HEAT EXCHANGER				Tag.No.:	
				PERFORMANCE DATA				CL7.1	
				PROJECT NAME: Nickel recovery from NI-MH batteries by H2SC				SHEET 1 of 2	
				PROJECT No.: 1				REV. 2	
				PLANT: 1				BY: Team 5	
								DATE: 9/6/24	
NAME:		Heat Exchanger CL7.1							
FUNCTION:		Cooling before reactor							
OPERATION:		Continuous							
MATERIAL HANDLED:		xxx							
PERFORMANCE OF ONE UNIT									
		IN		SHELL SIDE		OUT			
				cold air				black powder	
FLUID									
TOTAL FLOW	Kg/hr	3047		3047		625.6		625.6	
VAPOR	Kg/hr	3047		3047		0		0	
SOLID	Kg/hr	0		0		625.6		625.6	
NON CONDENSABLES	Kg/hr								
MOLECULAR WEIGHT		28.96		28.96		92.7		92.7	
DENSITY	Kg/m ³	1.268		0.947		3695.889		4061.269	
VISCOSITY	cP								
SPECIFIC HEAT	Kcal/Kg*K	0.24		0.241		0.1723		0.1723	
TEMPERATURE	°C	5		75		550		75	
OPERATING PRESS	barg	1		1		1		1	
VELOCITY	cp	0.0174		0.02066		*		*	
				CLEAN:		*		DESIGN:	
HEAT EXCHANGED: Mj/kg		-214.37		MTD(CORR)				EFF SURFACE m²:	
SURFACE PER UNIT: m ²		ALCULATED:		*		PROVIDED :		*	
SHELLS PER UNIT: No.		*		SIZE:		*		TYPE	
INSTALLED:				CONNECTED IN:					
STRUCUTION									
		GN PRESS: barg				DESIGN TEMP: °C		CORR.ALLOW: mm	
SHELL SIDE		1				150-700		N/A	
TUBE SIDE		1				0-100		N/A	
CODE REQUIREMENTS: TEMA CLASS		class "B" Type BEM							
SHELL DIAM/ID:		*		MAX.ALLOW.BUNDLE DIAM.(O.T.L):in		*			
BAFFLES TYPE:		*		SPACING: in		CUT/TYPE:		*	
LONGITUDINAL BAFFLE:		*		IMPINGE BAFFLE:		*			
TUBES:O.D. in		*		LENGTH: *		THICKNESS: *		PITCH: in *	
OF CONSTRUCTION									
SHELL		SS-316L				TUBE SHEET		SS-316L	
TUBES:O.D.		SS-316L				HEADS		SS-316L	
GASKETS									
Rev.	Description	Proc.Eng.	Date	Proc.Man.	Date	Proc.Eng.	Date	DSBG Appr.	Date
1	Heat exchanger CL7.1	Team 5	9/6/24						

T&S HEAT EXCHANGER					Tag.No.: CL7.1				
PERFORMANCE DATA									
PROJECT NAME:					Nickel recovery from NI-MH batteries by H2SC				
PROJECT No.:					1				
PLANT:					1				
SHEET					1 of 2				
REV.					2				
BY:					Team 5				
DATE:					9/6/24				

7.1m

3m

(1) (2) (3) (4)

MARK	DIAML	SERVICE	FLG.#	REMARKS
1	2"	black powder inlet	ANSI #150	
2	2"	black powder outlet	ANSI #150	
3	6"	inlet of cold air	ANSI #300	
4	6"	outlet of cold air	ANSI #150	

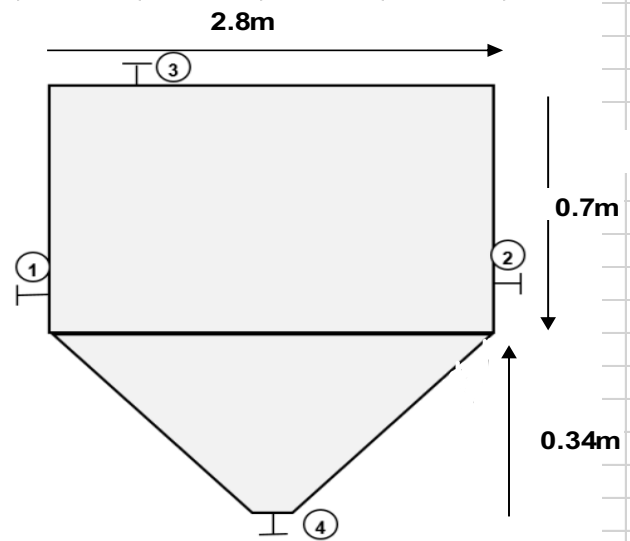
Rev.	Description	Proc.Eng.	Date	Proc.Man.	Date	Proc.Eng.	Date	DSBG Appr.	Date
1	Heat exchanger HE-02	Team 5	9/6/24						

				TANK/VESSEL				Tag.No.: CSTR-01			
				PERFORMANCE DATA							
				PROJECT NAME: Nickel recovery from batteries using H2SO4 leach				SHEET		1 of 2	
				PROJECT No.: 1				REV.		1	
				PLANT: 1				BY:		group5	
								DATE:		9/6/24	
NAME:		Reactor CSTR-1						MFS			
FUNCTION:		leaching process									
OPERATION:		Continuous									
MATERIALS											
PROCESS:		the leaching reaction occur in a CSTR reactor. A stream of H2SO4 acid and stream of battrey black powder enter to the reactor for occuring the leaching reaction that led to a nickel sulfate product.									
UTILITIES:											
PERFORMANCE INFORMATION											
		UNITS	NORMAL OPERATION				DESIGN BASIS				
WORKING VOLUME		lit	19600				20000				
FREEBOARD		lit	1400				1000				
TOTAL VOLUME		lit	21000				21000				
INTERNAL PRESSURE		barg	0-3				0-4				
INTERNAL TEMPERATURE		°C	50-90				30-120				
DENSITY		Kg/m ³	964.51				1000				
AREA OF JACKET/COIL		m ²	0.25				0.3				
HEATING OR											
COOLING FLUID		Kg/hr	9500				10000				
PRESSURE		barg	1				3				
TEMPERATURE		°C	20				50				
EQUIPMENT INFORMATION											
TYPE:		Fix Bed PFR				FLANGES:		ANSI #150			
						GASKETS:		Teflon enveloped			
DIMENSIONS:		Diam:	4000	mm	HEAT/COOL TYPE:		Water				
		High/Long:	5000	mm	AGITATOR:		NO				
MAT'L OF CONSTRUCTION						INSTRUMENT CONN.:		Yes			
HEADS:		SS-316L	CorrAllow:	2mm	VENT,PRESS.RELIEF:		Yes*				
SHELL:		SS-316L	CorrAllow:	2mm	CODE:		*				
SUPPORT:		SS-316L	CorrAllow:	2mm	INSULATION:		Yes (client responsibility)				
JACKET:						OTHERS:					
Rev.	Description	Proc.Eng.	Date	Proc.Man.	Date	Proc.Eng.	Date	DSBG Appr.	Date		
1	CSTR-01	group5	9/6/24								

				TANK/VESSEL		Tag.No.:			
				PERFORMANCE DATA		RC-01			
PROJECT NAME:				Nickel recovery from batteries using H ₂ SO ₄					
PROJECT No.:				1					
PLANT:				1					
				SHEET		2 of 2			
				REV.		1			
				BY:		group5			
				DATE:		9/6/24			
MARK	DIAM.	SERVICE	FLG.#	REMARKS					
A	3"	Product mixture outlet	ANSI #150						
B	1"	sulfuric acid inlet from the condenser	ANSI #150						
C	1"	reserve product mixture outlet	ANSI #150						
D	1"	black powder inlet	ANSI #150						
E	1/2"	Temp. Controller	ANSI #150						
F	1"	Cold Water Inlet	ANSI #150						
G	1"	Hot Water Outlet	ANSI #150						
H	1"	Drain	ANSI #150						
I	1/2"	pH Controller	ANSI #150						
J	1/2"	Level Controller	ANSI #150						
K	2"	sulfuric acid inlet	ANSI #150						
Rev.	Description	Proc.Eng.	Date	Proc.Man.	Date	Proc.Eng.	Date	DSBG Appr.	Date
1	CSTR-01	group5	9/6/24						

								Tag.No.:		
				Filter				VF-10		
				PERFORMANCE DATA						
				PROJECT NAME:				Nickel recovery from NI-MH batteries by H2SO4 leachir	SHEET	1 of 2
				PROJECT No.:				1	REV.	2
				PLANT:				1	BY:	Team 5
								DATE:	9/6/24	
NAME:		Filtration								
FUNCTION:										
OPERATION:		Continuous								
MATERIALS										
PROCESS:		Separate the precipitated NiSO4 crystals by filtration, retaining the NiSO4 crystals on the filter while CoSO4 and water pass through into filtrate								
UTILITIES:										
PERFORMANCE INFORMATION										
		UNITS	NORMAL OPERATION				DESIGN BASIS			
WORKING VOLUME		lit	5148				5200			
FREEBOARD		lit	1052				1000			
TOTAL VOLUME		lit	6200				6200			
INTERNAL PRESSURE		barg	0.05-1				0-1			
INTERNAL TEMPERATURE		°C	31.5				60			
DENSITY		Kg/m ³	2070				2100			
EQUIPMENT INFORMATION										
TYPE:		Horizontal				FLANGES:		ANSI#150		
						GASKETS:		EPDM		
DIMENSIONS:		Diam:	2800	mm		HEAT/COOL TYPE:		No		
		High/Long:	700	mm		AGITATOR:		No		
funnel		Diam:	2800	mm		INSTRUMENT CONN.:		Yes		
		High/Long:	340	mm		VENT,PRESS.RELIEF:		Yes		
MAT'L OF CONSTRUCTION						CODE:				
HEADS:		SS-316L	CorrAllow	1mm		INSULATION:		No		
SHELL:		SS-316L	CorrAllow	1mm		OTHERS:				
SUPPORT:		*	CorrAllow	1mm						
JACKET:			CorrAllow	1mm						
Rev.	Description	Proc.Eng.	Date	Proc.Man.	Date	Proc.Eng.	Date	DSBG Appr.	Date	
1	Filter VF-10	Team 5	9/6/24							

							Tag.No.:		
					FILTER		VF-10		
					PERFORMANCE DATA				
							SHEET	2 of 2	
				PROJECT NAME:	Nickel recovery from NI-MH batteries by H2SC			REV.	2
				PROJECT No.:	1			BY:	Team 5
				PLANT:	1			DATE:	9/6/24



MARK	DIAM.	SERVICE				FLG.#		REMARKS		
1	1"	black powder inlet				ANSI #150				
2	1"	NiSO4 outlet				ANSI #150				
3	1"	Vacuum				ANSI #150				
4	1"	CoSO4				ANSI #150				
Rev.	Description		Proc.Eng.	Date	Proc.Man.	Date	Proc.Eng.	Date	DSBG Appr.	Date
2	Filter VF-10		Team 5	9/6/24						

								Tag.No.:		
				TANK/VESSEL				ELE - 11		
				PERFORMANCE DATA						
						SHEET		1 of 2		
				PROJECT NAME:		Nickel recovery from batteries using H2SO4		REV. 1		
				PROJECT No.:		1		BY: group5		
				PLANT:		1		DATE: 9/6/24		
NAME:		electrolyses tank								
FUNCTION:		extract nickel metal from nickel sulfate crystal								
OPERATION:		Continuous								
MATERIALS										
PROCESS:		to extract a pure nickel metal from nickel sulfate crystal. A stream of nickel sulfate and stream of NaCl salt with a current inlet to ELE-11 for occurring the electrolysis reaction that led to a pure nickel product								
PERFORMANCE INFORMATION										
		UNITS	NORMAL OPERATION				DESIGN BASIS			
WORKING VOLUME		lit	3680				4000			
FREBOARD		lit	1320				1000			
TOTAL VOLUME		lit	5000				5000			
INTERNAL PRESSURE		barg	1				0-1.7			
INTERNAL TEMPERATURE		°C	31.5				80			
DENSITY		Kg/m ³	8900				9000			
EQUIPMENT INFORMATION										
TYPE: Vertical				FLANGES: ANSI #150						
				GASKETS: PTFE envelope						
DIMENSIONS:		Diam:	1000	mm	HEAT/COOL TYPE:		No			
		High/Long:	4000	mm	AGITATOR:		No			
MAT'L OF CONSTRUCTION				INSTRUMENT CONN.: Yes						
HEADS:		SS-316L	CorrAllow:	1mm	VENT,PRESS.RELIEF:		Yes*			
SHELL:		SS-316L	CorrAllow:	1mm	CODE:					
SUPPORT:		CS	CorrAllow:	1mm	INSULATION:		No			
JACKET:			CorrAllow:	1mm	OTHERS:					
Rev.	Description	Proc.Eng.	Date	Proc.Man.	Date	Proc.Eng.	Date	DSBG Appr.	Date	
1	ELE - 11	group5	9/6/24							

				TANK/VESSEL				Tag.No.:			
				PERFORMANCE DATA				ELE - 11			
PROJECT NAME:				Nickel recovery from batteries using H2SO4				SHEET	2 of 2		
PROJECT No.:				1				REV.	1		
PLANT:				1				BY:	group5		
								DATE:	9/6/24		
Diagram of Tank/Vessel ELE-11 showing dimensions (4.0m height, 1.00m diameter) and various ports (A, B, C, D, E, F) with associated valves and indicators.											
MARK	DIAM.	SERVICE			FLG.#		REMARKS				
E	1"	NaCl inlet to the tank ELE-11			ANSI #150						
C	1"	pure nickel metal outlet			ANSI #150						
D	1"	Electrolysis solution outlet			ANSI #150						
F	1"	Nickel Sulfate			ANSI #150						
B	1"	Voltage Indicator			ANSI #150						
A	1"	Temp Indicator			ANSI #150						
Rev.	Description			Proc.Eng.	Date	Proc.Man.	Date	Proc.Eng.	Date	DSBG Appr.	Date
1	ELE - 11			group5	9/6/24						

8. Description of system operation

8.1. System operation

The process we are planning is for nickel recovery from Ni-MH batteries using H₂SO₄ leaching. It is designed to produce 2950 tons of nickel metal per year.

Ni-MH batteries line:

The process begins once the Ni-MH batteries are removed from battery tank(T-1) for electrolysis using conveyor 0001. In electrolysis (DIS1), the batteries are discharged. The electrolysis system includes temperature and voltage indicators to monitor and maintain optimal electrolysis conditions. Afterward, the batteries are transferred to the shredding machine (CR2) via conveyor 0002, which is controlled by a flow control system (FC) that regulates the conveyor speed. In the shredding machine (CR2), the batteries are shredded into smaller pieces. The stream of shredded batteries is then transferred to a separator (DR3) via conveyor 0003, controlled by practical size distribution control (PDSC) that governs the speed of the conveyor. The separator (DR3) aims to separate the black powder from the iron case and nylon. It is controlled by a level control connected to valve LCV-0001 linked to pipe 0004, responsible for water inlet. Additionally, density control is performed by valve 0002-DCV connected to pipe 0006, managing the exit of iron cases with water from the separation tank. The flow of black powder exiting the separation tank is pumped through pump P-02, which increases the pressure from 1 atm to 1.4 atm in order to facilitate further processing. After pumping, the pipe enters the washing stage (WR4), where the black powder is washed with water to remove sodium and potassium. Following this stage, the black powder is transferred to the drying process using conveyor 0010, controlled by a density controller (DC) connected to valve DCV-0003 linked to pipe 0008. Subsequently, the black powder flows into a dryer (DR5) to dry it from 25[°C] to 80[°C] . After drying, the material is transferred to a furnace (CN6) via conveyor 0013, controlled by a moisture controller connected to valve MCV-0004, which is linked to pipe 0011. Conveyor 0013 enters the furnace (CN6) where the black powder undergoes combustion at 550[°C] to remove organic matter. The furnace includes a temperature indicator to control the temperature at 550 [°C] . The processed black powder exits the furnace with the assistance of conveyor 0016, controlled by an infrared controller (IRC) connected to valve IRCV-0005, which is linked to pipe 0014. After calcination, the black powder is transferred to cooler tank CL7.1 by a conveyor (0016), where it undergoes cooling to 75°C, essential for the subsequent leaching process. This cooling is regulated by a temperature control system connected to the TCV-0006 valve, which adjusts the flow rate of cold water entering CL7.1, ensuring uniform cooling. A flow indicator (0017) monitors the cold-water flow rate, providing real-time data to maintain optimal cooling performance. The heated water exits through pipe 0018 to a hot water treatment system. The cooled black powder is then continuously transferred from CL7.1 by conveyor (0019), also managed by the temperature control system to ensure only adequately cooled material proceeds. This integrated control system ensures precise temperature regulation, preventing overheating or undercooling, thereby maintaining the efficiency and consistency necessary for the leaching stage. After cooling to 75°C, the black powder is transferred to the Continuous Stirred-Tank Reactor (CSTR1), which is equipped with external cooling jackets, a magnetic stirrer, a sensor-based temperature controller, and a reflux condenser (COND8) to prevent evaporation losses.

The leaching reaction occurs at a pH of 1, temperature of 75°C, and ambient pressure. The reactor CSTR1's temperature is regulated by a control system connected to the TCV-0007 valve, which adjusts the cold-water flow rate into the cooling jackets, with heated water exiting through pipe 0022 to a hot water treatment system. Sulfuric acid (2M) is introduced into the reactor through pipe 0020 from T-2, controlled by a flow control and pH control system connected to the FCV-0008 valve, which regulates the acid flow rate based on the input black powder flow and pH measurements to maintain optimal leaching conditions (pH=1 and a solid-to-liquid ratio of 100 gr:1L). The leached black powder (with Nickel sulfate 12.8%wt) is continuously pumped from CSTR1 by pump P-04, increasing the pressure from 1 atm to 3 atm in pipe 0021, which is managed by pH and level controls connected to the LCV-0009 valve, ensuring proper flow rate adjustments based on real-time measurements of pH and reactor level.

After the leaching, the leached black powder is transferred to an evaporator (EVAP9) where it is heated to 100°C-150°C in an electric heater to evaporate most of the byproduct water, thus increasing the solute concentration, which is essential for the subsequent crystallizing process. The evaporator EVAP9 features real-time level and pressure indicators for safety considerations, and its electric heating element is controlled by a temperature control system to maintain the temperature between 100°C and 150°C. This ensures the efficient evaporation of water while preventing the evaporation of sulfuric acid, as the partial pressure of sulfuric acid in this temperature range is insignificant compared to that of water. The heated leached solution, now containing 46.5% wt nickel sulfate, is continuously transferred through pipe 0024 to the cooler (CL7.2) by pump P-05, which increases the pressure from 1 atm to 3 atm. The flow in pipe 0024 is regulated by a density control system that measures the concentration of nickel sulfate and adjusts the temperature of EVAP9 accordingly to keep it within the optimal range. The evaporated water exits through pipe 0025 and is directed to a water vapor treatment system for further processing.

After the evaporation, the heated leached solution (with a highly concentrated nickel sulfate solution of 46.5% wt) is continuously transferred to cooler CL7.2 for the crystallizing process. In this stage, the solution is supercooled to 31.5°C, causing the nickel sulfate to form crystals while cobalt sulfate remains in the solution due to its higher solubility. The cooler is equipped with temperature control to maintain optimal crystallizing conditions.

The resulting mixture, now containing nickel sulfate crystals, is transferred through pipe 0026 to vacuum filtration (VF-10) to separate the precipitated NiSO₄ crystals from the solution. This solution still contains dissolved cobalt sulfate (CoSO₄), water, dissolved nickel hydroxide, and sulfuric acid. The NiSO₄ crystals are retained on the filter, while the remaining solution exits the vacuum filtration via pipe 0028 with dissolved CoSO₄ (8.17% wt). The nickel sulfate crystals are conveyed by conveyor 0027 to the electrolysis unit (ELE-11). The conveyor system includes moisture control to measure and adjust the moisture level of the crystals. This moisture control is linked to the direction control of the conveyor and the MCV-0010 valve in pipe 0026, which regulates the flow rate of the NiSO₄ solution entering the vacuum filtration, thereby adjusting the direction of the conveyor and the flow rate of the NiSO₄ solution accordingly.

After the crystallization, the NiSO_4 crystals are transferred to ELE-11 to be processed through advanced electrolysis at 30°C . The electrolysis reaction carried out at an electrode potential of -0.26 volts versus the Standard Hydrogen Electrode (SHE), yields high-purity nickel metal. The electrolysis tank includes temperature and voltage indicators to monitor and maintain optimal electrolysis conditions. The tank contains three cells where the nickel metal forms, each processed sequentially in a batch process. After electrolysis, the nickel is moved by conveyor 0031 to the nickel storage tank (T-3) for shipment.

Conveyor 0031 is equipped with an XRD (X-ray diffraction) indicator to measure and ensure the required quality and purity of the produced nickel. Additionally, there is a flow control system in the electrolysis tank that measures the flow rate of the pure nickel, which helps monitor the required quantity and adjust the electrolysis voltage accordingly. Pipe 0029 supplies NaCl 15% wt into the electrolysis tank, while pipe 0030 removes the electrolysis solution, which contains 60% wt Na_2SO_4 , for further treatment.

H₂SO₄ acid:

The sulfuric acid (H_2SO_4) line originates from tank T-2, which stores a 2M H_2SO_4 solution. The acid is transported via pipe 0020 to the (CSTR1), where its flow is regulated by a flow control and pH control system connected to the FCV-0008 valve to maintain optimal leaching conditions ($\text{pH}=1$, solid-to-liquid ratio of 1:10). In CSTR1, the H_2SO_4 reacts with the black powder to form a nickel sulfate solution. Evaporation losses are prevented by reflux condenser COND8. The leached solution, containing H_2SO_4 (7.55% wt), is pumped from CSTR1 by pump P-04 through pipe 0021, with flow managed by pH and level controls connected to the LCV-0009 valve. It then enters the evaporator (EVAP9), where it is heated to 100°C - 150°C to evaporate most of the water and increase the concentration of solute. The concentrated solution (with sulfuric acid at 27.5% wt) is transferred via pipe 0024 to cooler CL7.2 by pump P-05. In cooler CL7.2, the solution is supercooled to 31.5°C . The mixture is subsequently transferred through pipe 0026 to vacuum filtration (VF-10), where NiSO_4 crystals are separated from the remaining solution containing dissolved CoSO_4 , nickel hydroxide, water, and sulfuric acid at 51.42% wt. The filtrate exits through pipe 0028 for further treatment.

8.2. Alerts

Discharger (DIS-1)

Valve table:

Valve name	General status
0002	ON

Alerts table:

Serial No.	Alert name	Limit for alert activation	Operations when alert activates	Conditions for exiting alert
1	High flow	FM-DIS1 > 0.4 m ³ /hr for 30 sec	1. Alert sent to operator on control screen 2. Change direction backward of the Conveyor 0002	1. FM-DIS1 < 0.37 m ³ /hr for 30 sec 2. Approval of control operator for exiting the alert
2	Low flow	FM-DIS1 < 0.3 m ³ /hr for 30 sec	1. Alert sent to operator on control screen 2. Move forward 0002	1. FM-DIS1 > 0.33 m ³ /hr for 30 sec 2. Approval of control operator for exiting the alert

Crusher (CR-2)

Valve table:

Valve name	General status
0003	ON

Alerts table:

Serial No.	Alert name	Limit for alert activation	Operations when alert activates	Conditions for exiting alert
1	High Particle size	PDSM-CR2> 0.1mm for 30 sec	1. Alert sent to operator on control screen 2. Slightly close 0003	1. PDSM-CR2< 0.7 mm 2. Approval of control operator for exiting the alert
2	Low Particle size	PDSM-CR2< 0.5mm for 30 sec	1. Alert sent to operator on control screen 2. Slightly open 0003	1. PDSM-CR2> 0.5 mm 2. Approval of control operator for exiting the alert

Density separation (DR3)

Valve table:

Valve name	General status
LCV-0001	ON
DCV-0002	ON

Serial No.	Alert name	Limit for alert activation	Operations when alert activates	Conditions for exiting alert
1	Low density	DM-DR3< 2010 kg/m ³ for 30 sec	1. Alert sent to operator on control screen 2. Slightly close DCV-0002	1. Dm-0005> 2005 kg/m³ for 30 sec 2. Approval of control operator for exiting alert
2	High density	DM-DR3> 1990 kg/m ³ for 30 sec	1. Alert sent to operator on control screen 2. Slightly open DCV-0002	1. Dm-0005< 1995 kg/m³ for 30 sec 2. Approval of control operator for exiting alert
3	High level	LM- DR3> 95% for 10 sec	1. Alert on control screen “DR3 level too high” 2. Slightly close LCV-0001	1. LM- DR3< 95% for 10 sec 2. Approval of control operator
4	low level	LM- DR3< 75% for 10 sec	1. Alert on control screen “DR3 level too low” 2. Fully open LCV-0001	1. LM- DR3> 75% for 10 sec 2. Approval of control operator

Washer (WR-4)

Valve table:

Valve name	General status
DCV-0003	ON

Alerts table:

Serial No.	Alert name	Limit for alert activation	Operations when alert activates	Conditions for exiting alert
1	Low density	Dm-0010 < 3010 kg/m ³ for 30 sec	1. Alert sent to operator on control screen 2. Slightly open DCV-0003	1. Dm-0010 > 3015 kg/m ³ for 30 sec 2. Approval of control operator for exiting alert
2	High density	Dm-0010 > 3030 kg/m ³ for 30 sec	1. Alert sent to operator on control screen 2. Slightly close DCV-0003	1. Dm-0010 < 3025 kg/m ³ for 30 sec 2. Approval of control operator for exiting alert

Drayer (DR-5)

Valve table:

Valve name	General status
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MCV-0004	ON
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Alerts table:

Serial No.	Alert name	Limit for alert activation	Operations when alert activates	Conditions for exiting alert
1	High Moisture	MM-0013> 15% for 30 sec	- Alert sent to operator on control screen - Fully open MCV-0004	- MM-0013< 5% for 30 sec - Approval of control operator
2	Low Moisture	MM-0013< 5% for 30 sec	Alert sent to operator on control screen	- Dm-0008< 1410 kg/m³ for 60 sec - Approval of control operator

Calcination (CN-6)

Valve table:

Valve name	General status
IRCV-0005	ON

Alerts table:

Serial No.	Alert name	Limit for alert activation	Operations when alert activates	Conditions for exiting alert
1	High organic material concentration	IRM-0015 indicate a high organic material concentration	1. Alert sent to operator on control screen 2. Slightly open IRCV-0005	1. IRM-0015 indicate a low organic material concentration. 2. Approval of control operator
2	Low organic material concentration	IRM-0015 indicate a low organic material concentration	1. Alert sent to operator on control screen 2. Slightly close IRCV-0005	1. Approval of control operator

Cooler1 (CL-7.1)

Valve table:

Valve name	General status
TCV-0006	ON

Alerts table:

Serial No.	Alert name	Limit for alert activation	Operations when alert activates	Conditions for exiting alert
1	High temperature	TM-0019> 80 °C for 10 sec	1. Slightly open TCV-0006 2. Alert on control screen “CL7.1 temperature too high”	1. TM-0019< 77°C for 10 sec 2. Approval of control operator
2	Low temperature	TM-0019< 70°C for 10 sec	1. Slightly close TCV-0006 2. Alert on control screen “CL7.1 temperature too low”	1. TM-0019> 73°C for 10 sec 2. Approval of control operator

CSTR reactor (CSTR-1)

Valves & motors table:

Valve name	General status
TCV-0007	ON
DCV-0008	ON
LCV-0009	ON

Alerts table:

Serial No.	Alert name	Limit for alert activation	Operations when alert activates	Conditions for exiting alert
1	High temperature	TM- CSTR-1 > 80 °C for 30 sec	1. Slightly open TCV-0007 2. Alert on control screen “CSTR-1 temperature too high”	1. TM- CSTR-1 < 77°C for 30 sec 2. Approval of control operator
2	Low temperature	TM- CSTR-1 < 70°C for 30 sec	1. Slightly close TCV-0007 2. Alert on control screen “CSTR-1 temperature too low”	1. TM- CSTR-1 > 73°C for 30 sec 2. Approval of control operator
3	High Flow	FM-0019 > 0.20 m ³ /hr for 30 sec	1. Alert sent to operator on control screen	1. FM-0019 < 0.18m ³ /hr for 30 sec

			2. Slightly open DCV-0008	2. Approval of control operator
4	Low Flow	FM-0019< 0.10 m ³ /hr for 30 sec	1. Alert sent to operator on control screen 2. Slightly close DCV-0008	1. FM-0019< 0.13m ³ /hr for 30 sec 2. Approval of control operator
5	High level	LM- CSTR-1> 95% for 10 sec	1. Alert on control screen “CSTR-1 level too high” 2. Slightly close LCV-0009	1. LM- CSTR-1< 90% for 10 sec 2. Approval of control operator
6	low level	LM- CSTR-1< 75% for 10 sec	1. Alert on control screen “CSTR-1 level too low” 2. Slightly open LCV-0009	1. LM- CSTR-1> 80% for 10 sec 2. Approval of control operator
7	High pH	pHM-CSTR-1<0.8 for 30 sec	1. Alert on control screen “CSTR-1 pH too High” 2. Slightly close DCV-0008	1. pHM-CSTR-1>1 for 30 sec 2. Approval of control operator
8	low pH	pHM-CSTR-1>2 for 30 sec	1. Alert on control screen “CSTR-1 pH too low” 2. Slightly open DCV-0008	1. pHM-CSTR-1<1.5 for 30 sec

				2. Approval of control operator
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Evaporator (EVAP-9)

Valves table:

Valve name	General status
0024	ON
LCV-0010	ON

Alerts table:

Serial No.	Alert name	Limit for alert activation	Operations when alert activates	Conditions for exiting alert
1	High density	DM-0024 > 880 kg/m ³ for 30 sec	1. Alert sent to operator on control screen 2. Slightly close 0024	1. DM-0024 < 875 kg/m ³ for 30 sec 2. Approval of control operator for exiting alert
2	Low density	DM-0024 < 870 kg/m ³ for 30 sec	1. Alert sent to operator on control screen 2. Slightly open 0024	1. DM-0024 > 872 kg/m ³ for 30 sec

				2. Approval of control operator for exiting alert
3	High temperature	TM-EVAP-9> 110 °C for 30 sec	1. Slightly open 0024 2. Alert on control screen “EVAP-9 temperature too high”	1. TM-EVAP-9<105°C for 30 sec 2. Approval of control operator
4	Low temperature	TM-EVAP-9< 90°C for 30 sec	1. Slightly close 0024 2. Alert on control screen “EVAP-9 temperature too low”	1. TM-EVAP-9>95°C for 30 sec 2. Approval of control operator
5	High level	LM- EVAP-9> 95% for 10 sec	3. Alert on control screen “EVAP-9 level too high” 1. Slightly close LCV-0010	3. LM- EVAP-9 < 90% for 10 sec 1. Approval of control operator
6	low level	LM- EVAP-9< 75% for 10 sec	3. Alert on control screen “EVAP-9 level too low” 4. Slightly open LCV-0010 2.	3. LM- EVAP-9> 80% for 10 sec 2. Approval of control operator

Cooler2 (CL-7.2)

Valves table:

Valve name	General status
0024	ON

Alerts table:

Serial No.	Alert name	Limit for alert activation	Operations when alert activates	Conditions for exiting alert
1	High temperature	TM-CLE-7.2> 35 °C for 10 sec	1. Slightly close 0024 2. Alert on control screen “CL7.2 temperature too high”	1. TM-CLE-7.2< 33°C for 10 sec 2. Approval of control operator
2	Low temperature	TM-CLE-7.2< 25°C for 10 sec	3. Slightly open 0024 4. Alert on control screen “CL7.2 temperature too low”	3. TM-0019> 28°C for 10 sec 4. Approval of control operator

Vacuum Filter (VF-10)

Valves table:

Valve name	General status
MCV-0011	ON

Alerts table:

Serial No.	Alert name	Limit for alert activation	Operations when alert activates	Conditions for exiting alert
1	High Moisture	MM-0028 > 10% for 30 sec	1. Alert sent to operator on control screen 2. Slightly close MCV-0010.	1. MM-0028 < 3% for 30 sec 2. Approval of control operator
2	Low Moisture	MM-0013 < 5% for 30 sec	1. Fully open MCV-0010.	

Electrolysis Tank (ELE-11)

Valves table:

Valve name	General status
0031	ON

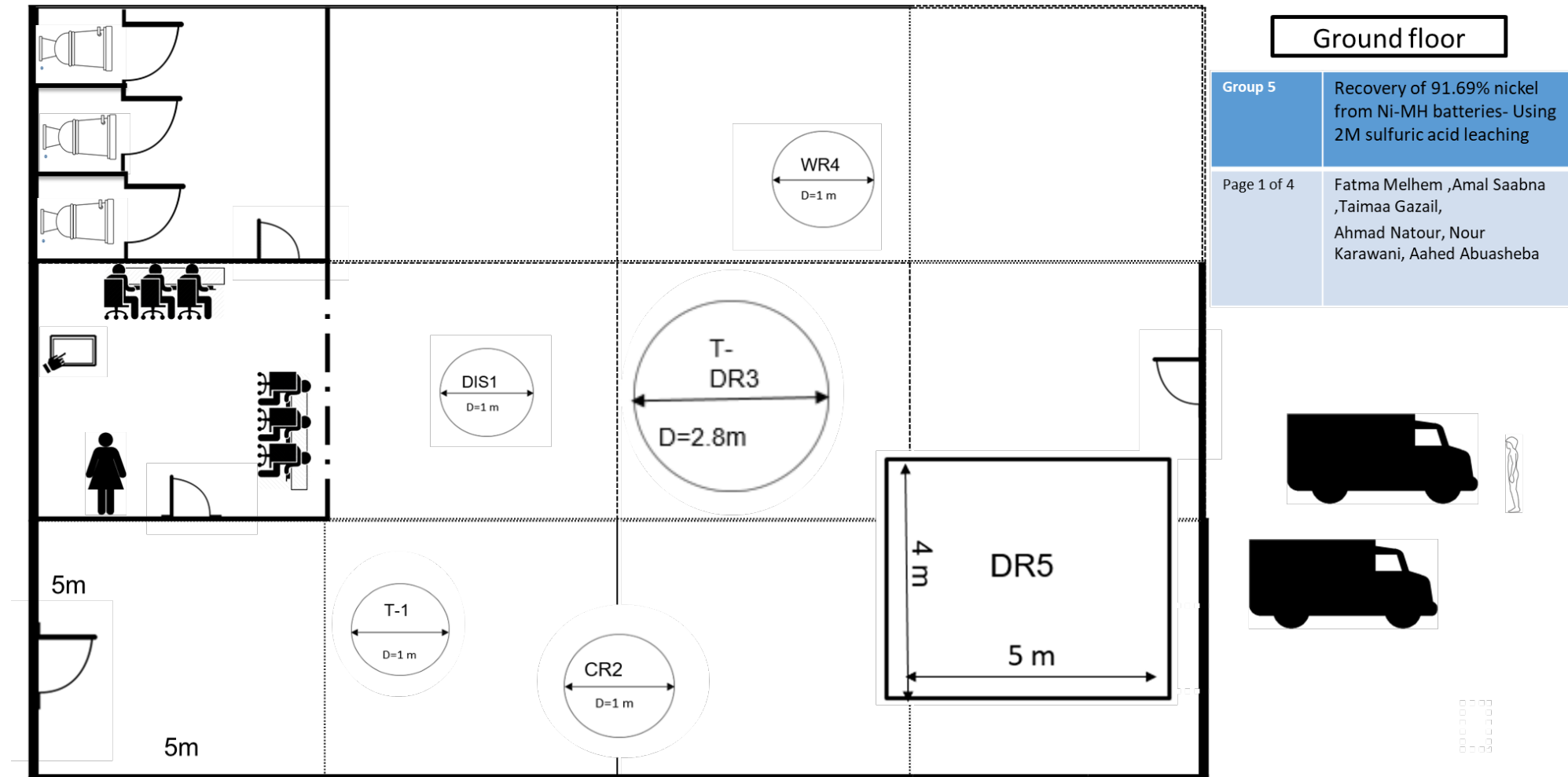
Alerts table:

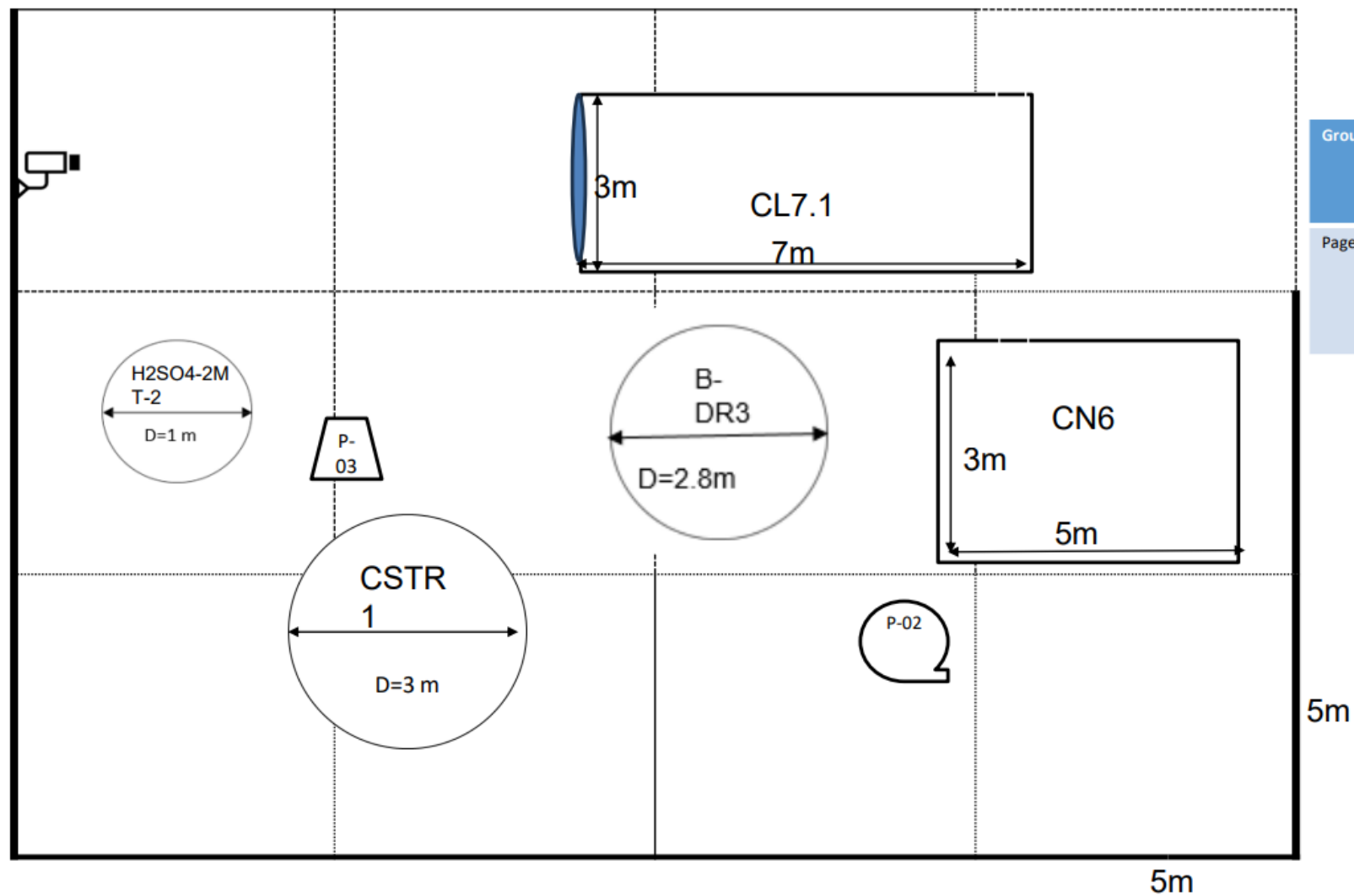
Serial No.	Alert name	Limit for alert activation	Operations when alert activates	Conditions for exiting alert

1	High Flow	FM-0031 > 0.045 m ³ /hr for 30 sec	1. Alert sent to operator on control screen 2. Slightly close 0031 3. increase in voltage for electrolysis	1. FM-0028 < 0.04m³/hr for 30 sec 2. Approval of control operator
2	Low Flow	FM-0031 < 0.03 m ³ /hr for 30 sec	1. Alert sent to operator on control screen 2. Slightly open 0031 3. decrease in voltage for electrolysis	1. FM-0028 < 0.035m³/hr for 30 sec 2. Approval of control operator

9. Equipment systems

9.1 Superior view

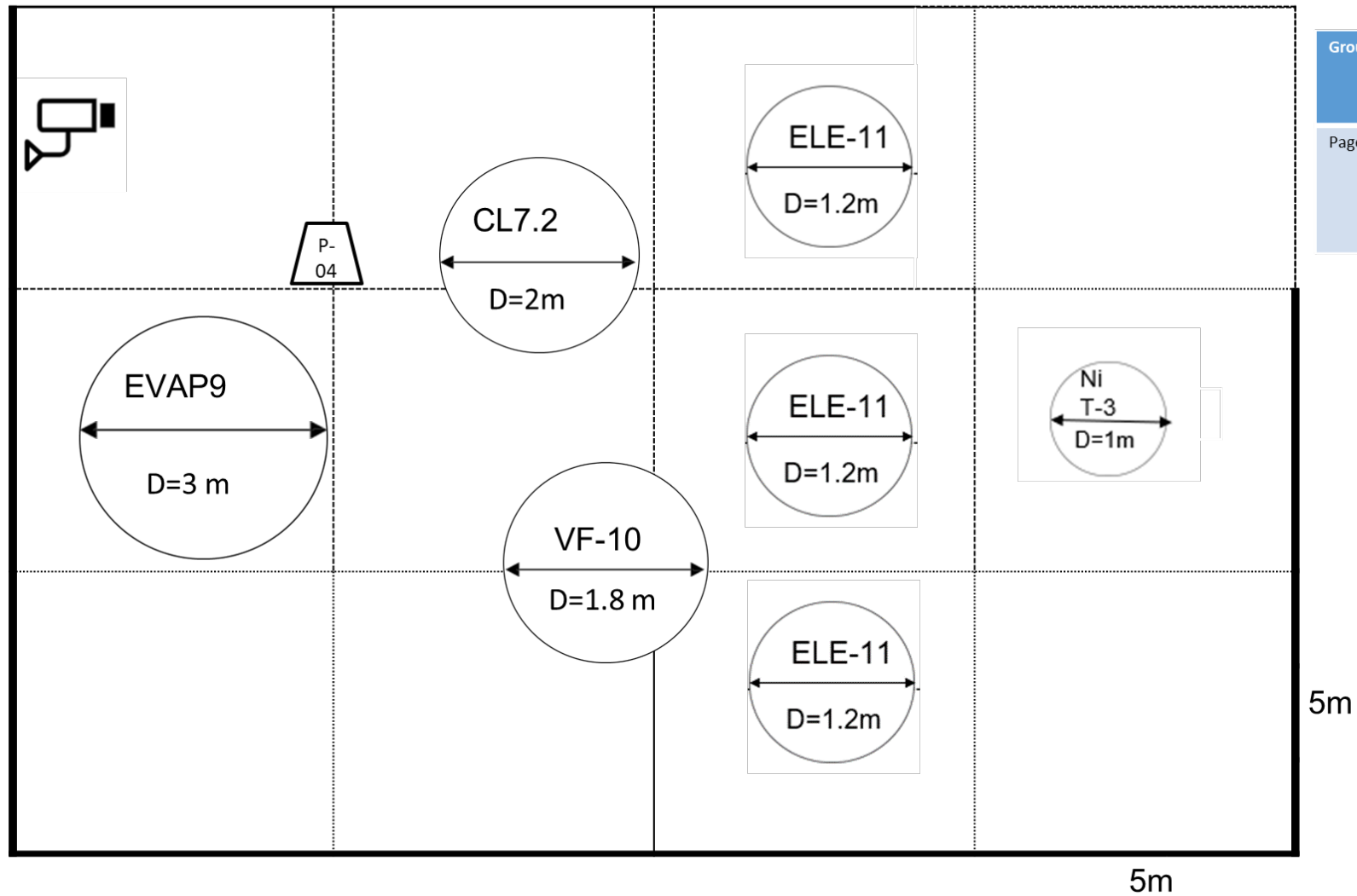




First floor

Group 5	Recovery of 91.69% nickel from Ni-MH batteries- Using 2M sulfuric acid leaching
Page 2 of 4	Fatma Melhem ,Amal Saabna ,Taimaa Gazail, Ahmad Natour, Nour Karawani, Aahed Abuasheba

Second floor

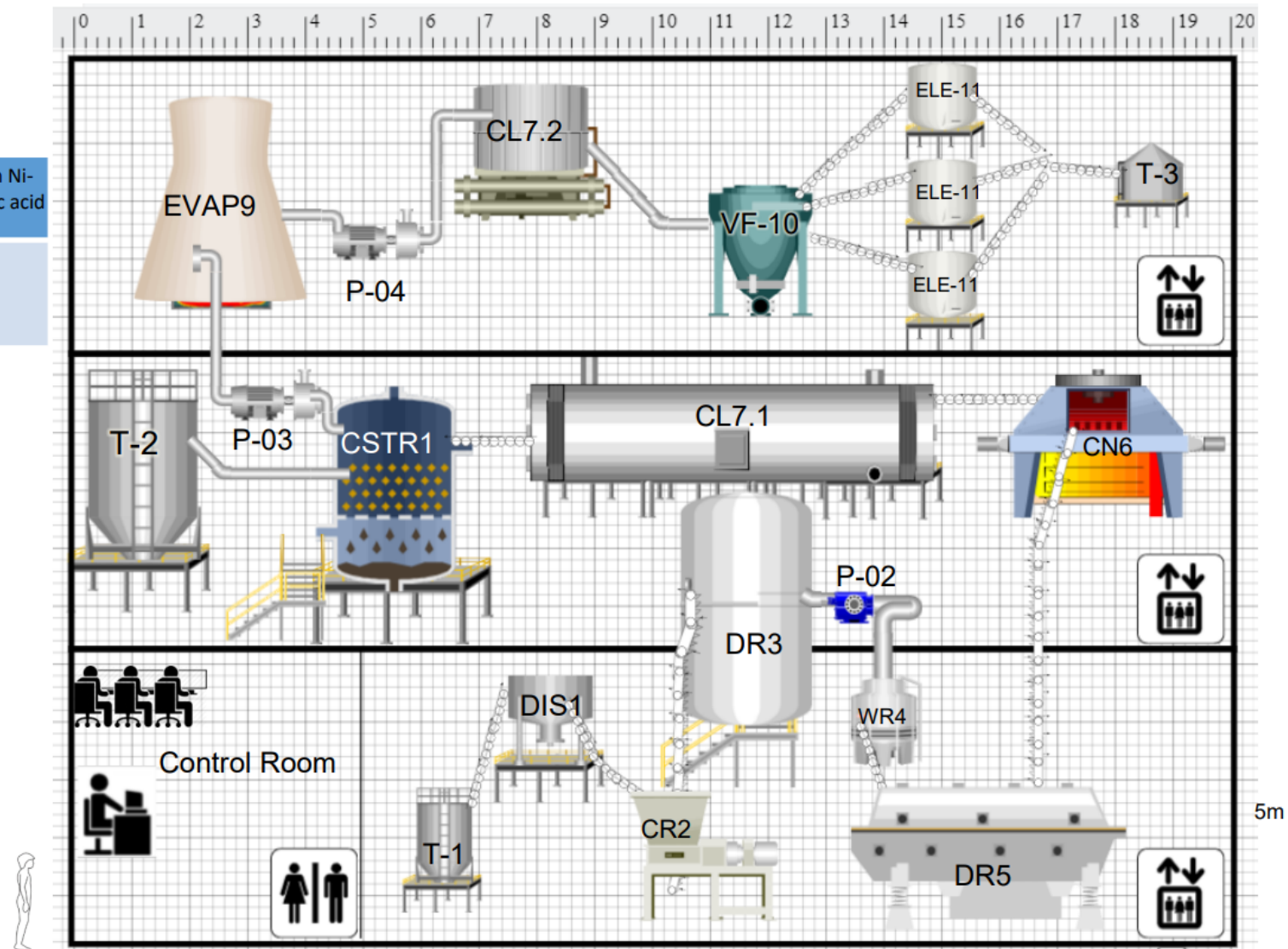


Group 5	Recovery of 91.69% nickel from Ni-MH batteries- Using 2M sulfuric acid leaching
Page 3 of 4	Fatma Melhem ,Amal Saabna ,Taimaa Gazail, Ahmad Natour, Nour Karawani, Aahed Abuasheba

9.2. Side View

Side view

Group 5	Recovery of 91.69% nickel from Ni-MH batteries- Using 2M sulfuric acid leaching
Page 4 of 4	Fatma Melhem ,Amal Saabna ,Taimaa Gazail, Ahmad Natour, Nour Karawani, Aahed Abuasheba



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11. Appendices:

11.1 Example calculation for heat balance on the dryer- DR5:

$$m_{in} \cdot Cp_{in} \cdot \Delta T_{10} = m_{out} \cdot Cp_{out} \cdot \Delta T_{11}$$

$m_{in} = 962 [\text{Kg/hr}]$ - The mass of the heated material, which is the black powder entering the dryer.

$Cp_{in} = 1.6577 [\text{KJ/Kg} \cdot \text{K}]$ - The heat capacity of the stream entering the dryer.

$Cp_{out} = 1.011 [\text{KJ/Kg} \cdot \text{K}]$ - The heat capacity of the incoming air to the dryer.

$\Delta T_{10} = T_{out} - T_{in} = 353[\text{K}] - 298[\text{K}]$ - The inlet and outlet temperatures of the black powder stream.

$T_{out,11} = 353[\text{K}]$ - The exit temperature of the air stream exiting the dryer.

Substitute into the equation and we get:

$$962 [\text{Kg/hr}] \cdot 1.6577 [\text{KJ/Kg} \cdot \text{K}] \cdot (353[\text{K}] - 298[\text{K}]) = m_{out} \cdot 1.011 [\text{KJ/Kg} \cdot \text{K}] \cdot (T_{in} - 353[\text{K}])$$

From the equation, it is evident that there are two unknowns. Assuming a relative humidity of 17% and a dew point temperature of 80[°C], the absolute humidity can be determined from the Psychrometric chart for air conditioning

$$0.022 = \frac{\text{Kg water}}{\text{Kg air}} = \frac{192 [\text{Kg}]}{\text{Kg air}}$$

$$m_{air} = 8680 [\text{Kg}]$$

We now have an equation with one variable, which is the inlet temperature of the air:

$$962 [\text{Kg/hr}] \cdot 1.6577 [\text{KJ/Kg} \cdot \text{K}] \cdot (353[\text{K}] - 298[\text{K}]) = 8680 [\text{Kg}] \cdot 1.011 [\text{KJ/Kg} \cdot \text{K}] \cdot (T_{in} - 353[\text{K}])$$

$$T_{in} = 363 [\text{K}]$$

It can be substituted into one of the balances to find the heat of the system:

$$Q = 962 \left[\frac{\text{Kg}}{\text{hr}} \right] \cdot 1.6577 \left[\frac{\text{KJ}}{\text{Kg}} \cdot \text{K} \right] \cdot (353[\text{K}] - 298[\text{K}])$$

$$Q = 87754 \left[\frac{\text{KJ}}{\text{hr}} \right]$$

11.2 Example calculation for mass balance on the evaporator-EVAP9:

$$m_{In} = \sum m_{out}$$

$$m_{In} = m_{out(22)} + m_{out(23)}$$

$m_{in} = 6894.9 \text{ [Kg/hr]}$ - The mass entering the evaporator -stream No. 21.

$m_{out(22)}$ The mass of the liquid exiting the evaporator -stream No. 22.

$m_{out(23)} = 5000 \text{ [Kg/hr]}$ The desired mass of the water vapor exiting the evaporator -stream No. 23.

Substitute into the equation, and we get an equation with one variable:

$$6894.9 \left[\frac{\text{Kg}}{\text{hr}} \right] = m_{out(22)} + 5000 \left[\frac{\text{Kg}}{\text{hr}} \right]$$

$$m_{out(22)} = 6894.9 \left[\frac{\text{Kg}}{\text{hr}} \right] - 5000 \left[\frac{\text{Kg}}{\text{hr}} \right]$$

$$m_{out(22)} = 1894.9 \left[\frac{\text{Kg}}{\text{hr}} \right]$$

11.3 Example calculation for Pipe diameter:

$$A = \frac{\pi * D^2}{4} = \frac{Q}{V}$$

$A[m^2]$ - Area

$D[m]$ - Pipe diameter

$Q \frac{m^3}{sec}$ - Flow

$V \frac{m}{sec}$ - Fluid velocity

Assumption:

$$V_{gas} = 10 - 50 \frac{m}{sec}$$

$$V_{liquid} = 1 - 5 \frac{m}{sec}$$

$$\rightarrow D = \sqrt{\frac{4 * Q}{\pi * V}}$$

Example For gas Pipe :

$$V_{gas} = 10 \frac{m}{sec}$$

$$Q = 109.69 \frac{m^3}{hr} = 0.030 \frac{m^3}{sec}$$

$$D = \sqrt{\frac{4 * 0.030 \frac{m^3}{sec}}{\pi * 10 \frac{m}{sec}}} \cong 0.062 \text{ m} = 2.44" \rightarrow 3.00"$$

Example For *iquid* Pipe:

$$V_{liquid} = 1 \frac{m}{sec}$$

$$Q = 7.28 \frac{m^3}{hr} = 2.02 * 10^{-3} \frac{m^3}{sec}$$

$$D = \sqrt{\frac{4 * 2.02 * 10^{-3} \frac{m^3}{sec}}{\pi * 1 \frac{m}{sec}}} \cong 0.051 m = 2.007" \rightarrow 2.00"$$